

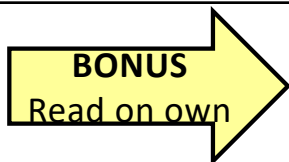
Geriatric Medicine

Dr. Shannon Gui

Sunday, November 30, 2025

Outline

- | | | |
|-------|--|-----------------|
| I. | Dementia and MCI | MIND |
| a. | Diagnosis | |
| b. | Management | |
| II. | Delirium | |
| III. | Depression | |
| IV. | Falls | MOBILITY |
| V. | Disease Management in Older Persons | MEDS |
| VI. | Polypharmacy | MULTICOMPLEXITY |
| VII. | Vaccinations in Older Persons | |
| VIII. | BONUS SLIDES: COVID and geriatrics, more disease management, normal aging, polypharmacy, weight loss, incontinence, constipation, bonus MCQs | |



Short forms used in this talk

Short	Full text
CAM	Confusion Assessment Method
MMSE	Mini Mental Status Exam (Folstein)
MoCA	Montreal Cognitive Assessment
RUDAS	Rowland Universal Dementia Assessment Scale
ADLs / IADLs	Activities of Daily Living / Instrumental Activities of Daily Living
SCI	Subjective Cognitive Impairment
MCI	Mild Cognitive Impairment
AD	Alzheimer's Dementia
VaD	Vascular Dementia
DLB	Dementia with Lewy Bodies
FTD	Frontotemporal Dementia

Short	Full text
EPS	Extra-pyramidal symptoms
PPA	Primary Progressive Aphasia
PCA	Posterior Cortical Atrophy
BPSD	Behavioural and Psychological Symptoms of Dementia
ChEI	Acetylcholinesterase Inhibitor (e.g. Donepezil, Galantamine, Rivastigmine)
NMDA- RA	N-methyl-D-aspartate Receptor Antagonist (e.g. Memantine)
SSRI	Selective Serotonin Reuptake Inhibitor
Others	DDx = Differential Diagnosis, Hx = History, P/E = Physical Exam, Rx = Treatment

Ia. Dementia & Mild Cognitive Impairment: Diagnosis

“Chronic brain failure”



Lightspring/Shutterstock

Key Guidelines:

1. *Lancet* Commission report on Dementia prevention, intervention, and care (2024)
2. WHO Guidelines on Risk reduction of cognitive decline and dementia (2019)
3. Recommendations of the 5th Canadian Consensus Conference on the diagnosis and treatment of dementia (2020)
4. Canadian Stroke Best Practice Recommendations: Vascular cognitive impairment guidelines update (2025)
5. Canadian Clinical Practice Guidelines for Assessing and Managing Behavioural and Psychological Symptoms of Dementia (2024)
6. CMA Driver's Guide 10th Edition, Section 8 (2023)

MCQ #1 2026

An 81-year-old female is evaluated for gradually worsening memory and functional decline over the past 3 years. Her daughter reports that she struggles with misplacing her wallet, paying bills and planning meals. She becomes irritable when the daughter tries to intervene and this has led to several verbal arguments between them. Her medical history is notable for HTN, T2DM, dyslipidemia, peripheral neuropathy, and GERD. Medications include metformin, empagliflozin, rosuvastatin, ramipril, pantoprazole. She sometimes appears slowed in thought and movement, and his family has noticed that her walking speed is reduced. On cognitive testing, she has difficulty with clock drawing and mini Trails B. Which of the following is the most likely diagnosis?

- A. Major depressive disorder
- B. Frontotemporal dementia
- C. Vascular dementia
- D. Alzheimer disease

Six Cognitive Domains

Executive function

*Problems with putting together a complex meal/hosting a family event?
Keeping track of bills?*

Personality

Lost interest in previous activities?

Attention

Difficulty following along TV shows? “Hearing difficulties” with normal audiology results

Language

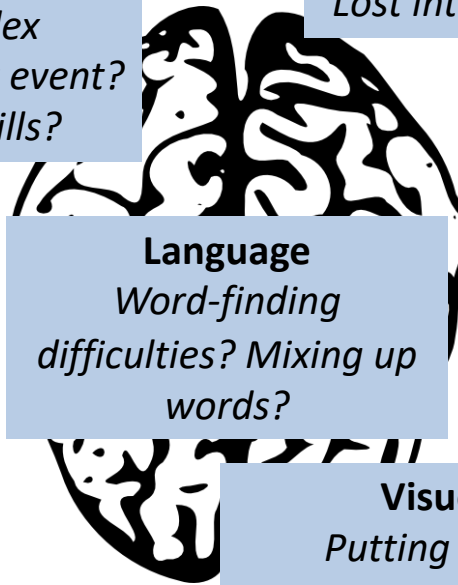
Word-finding difficulties? Mixing up words?

Memory

Repetitive questions, remembering appointments or losing things?

Visuospatial

Putting clothing on incorrectly? Gotten lost recently?





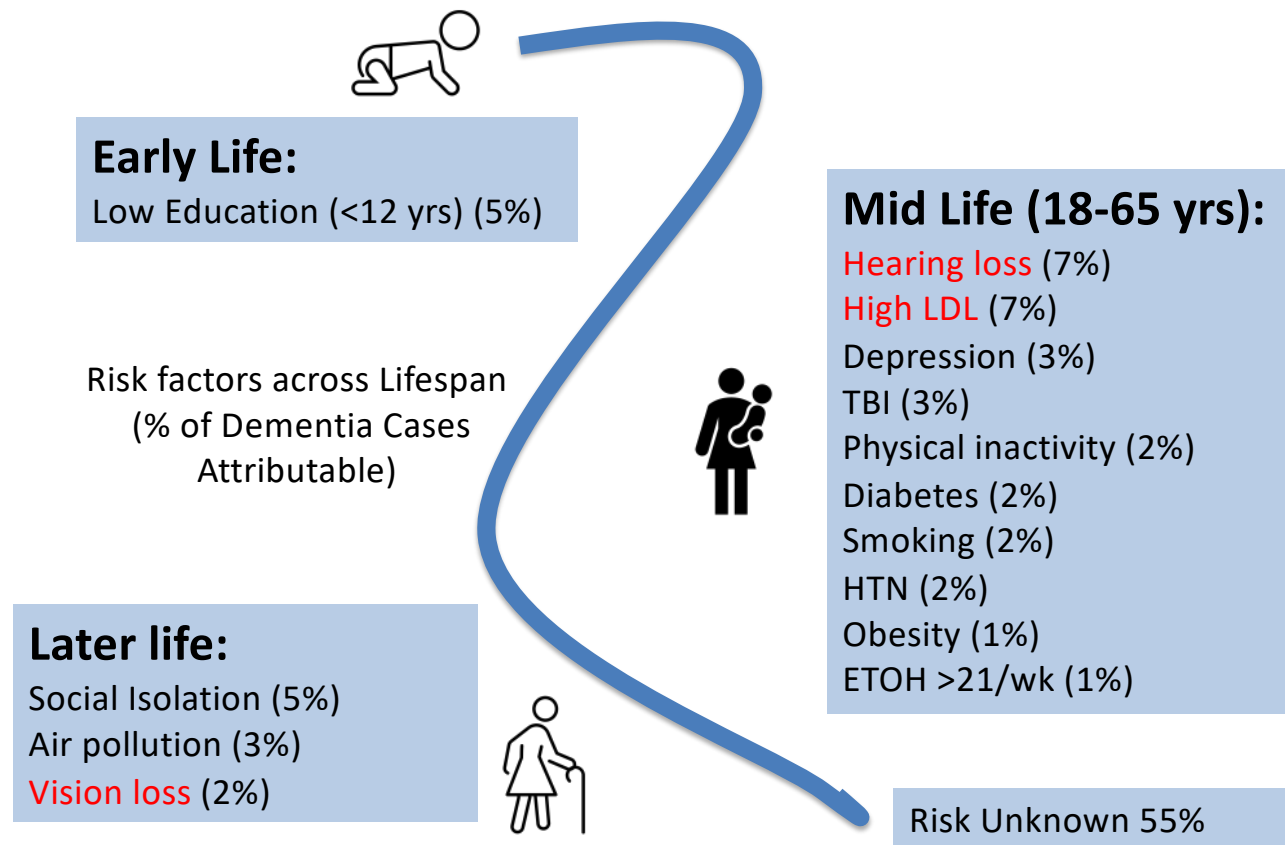
Normal aging

Decline in processing speed, performance speed, reaction time, fluid intelligence (flexible reasoning and problem-solving)

Normal aging is like an old computer with a full hard drive – everything is there, but it's hard to put in new data and slower to retrieve

	Preserved	Declines
Memory	Remote (“where were you born?”), procedural (ride a bike), general knowledge (“what is the capital of Ireland?”)	Working memory (reverse digit span), learning new info, free recall (i.e. “memorize this list of words”)
Language	Comprehension, vocabulary, phonemic fluency (“list words that begin with B”), grammar	Spontaneous word finding (cueable!), semantic fluency (“list animals”)
Attention	Sustained attention	Divided attention
Visuospatial	Simple copy	Complex copy, mental rotation

45% of dementia cases are preventable



Adapted from
Lancet. 2024;
404(10452): 572-628

Subjective Cognitive Impairment (SCI)

- Subjective cognitive decline
 - ? Age-related decline in short term/working memory
 - **No objective impairment** on cognitive testing
 - **Preserved function**
- Increased risk MCI and dementia
 - If corroborated by collateral history, consider yearly follow-up with cognitive testing
- Area of ongoing research

Do NOT screen asymptomatic older adults for cognitive impairment

Mild Cognitive Impairment (MCI)



a.k.a. Minor Neurocognitive Disorder (DSM-V)

- Concern regarding a change in cognition.
 - **Objective impairment** in 1 or more domains on cognitive testing
 - **Preserved function (normal IADLs/ADLs)** Can have coping strategies like lists, calendars, etc. but should not be a major change from baseline
- Not attributed to delirium, depression, psychosis, other medical/mental health disorder

Cognitive impairment with
normal function

Dementia

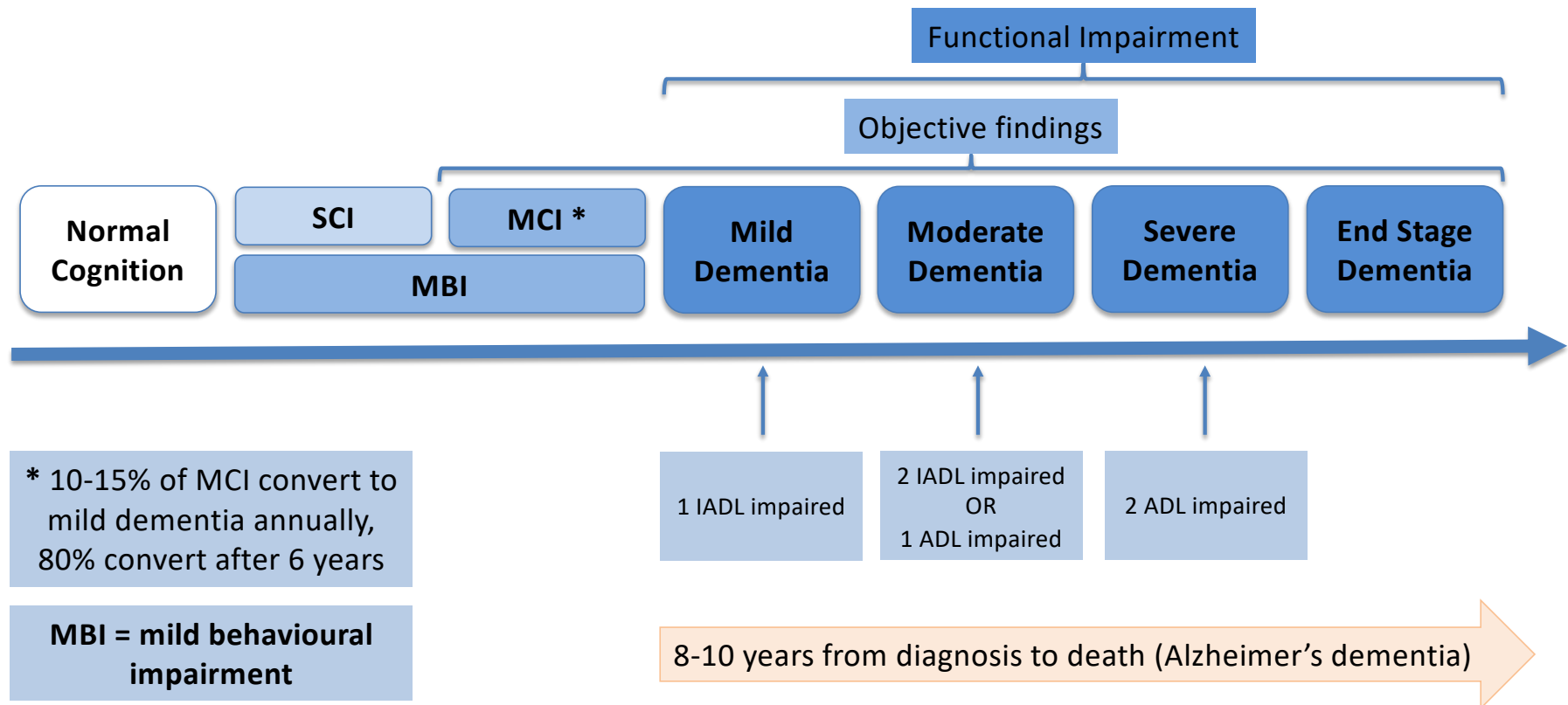


a.k.a. Major Neurocognitive Disorder (DSM-V)

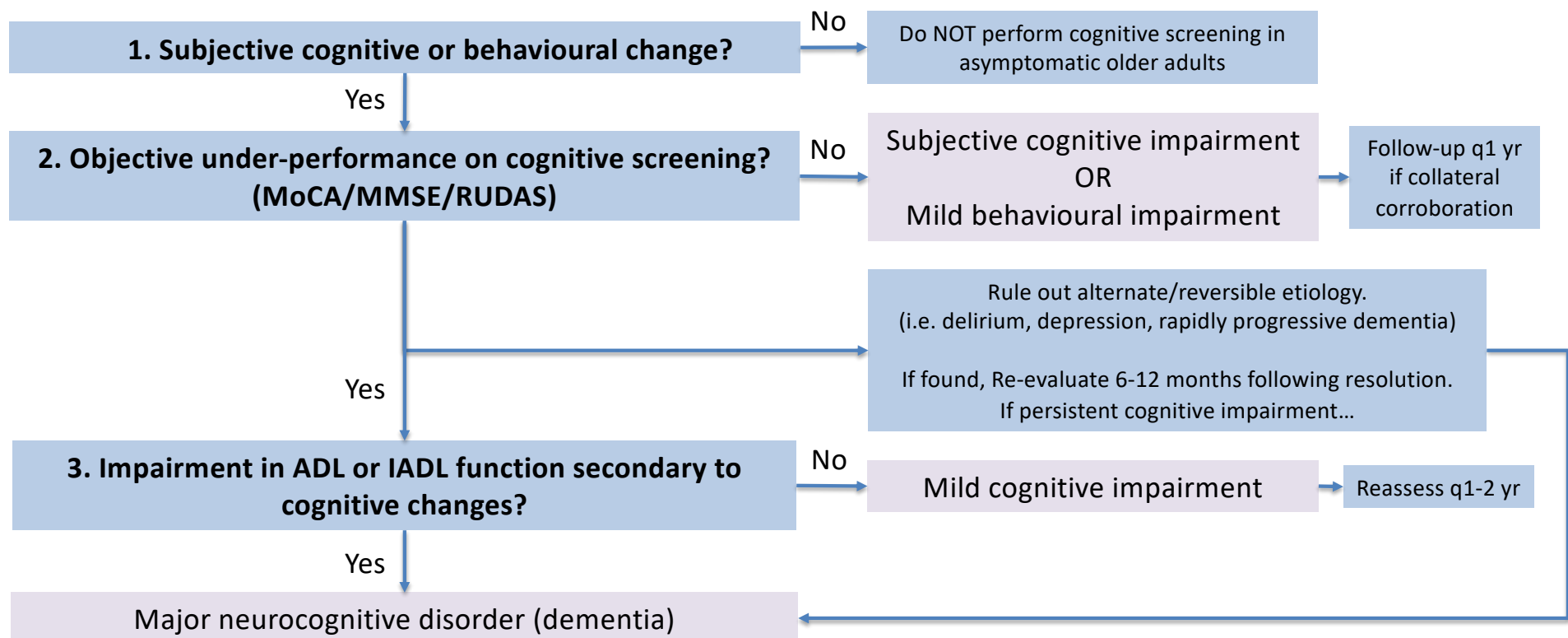
- Cognitive decline in 1 or more domains
 - **Objective impairment** on cognitive screen or testing
 - **Functional impairment (at least 1 IADL)** Earliest to go: driving, finances, medications, meal preparation¹
- Not attributed to delirium, depression, psychosis, other medical/mental health disorder

Cognitive impairment with
impaired function

Spectrum of cognitive decline



Approach to cognitive impairment



Briefly review these tests



Cognitive Testing

Cognitive Domain	MMSE <24/30: sensitivity (Sn) 85% for detecting <u>dementia</u>	MoCA <26/30: Sn >90 for detecting <u>MCI/mild dementia</u>	RUDAS <23/30: Sn 77% for detecting <u>dementia</u>
Executive Function	No	Trails, Clock draw, Abstraction, Letter fluency	Praxis sequence (complex), Crossing road
Language	Naming, Phrase repetition	Naming, Sentence repetition	Animal fluency
Attention	Serial 7s or WORLD	Digit span, Vigilance A, Serial 7s	No
Visuospatial	Intersecting pentagons	Trails, Cube, Clock draw	Body orientation, Cube
Immediate recall	Yes	Yes	Yes
Delayed recall	Yes	Yes	Yes
When to use	English-speaking, limited education	English-speaking, highly educated	Non-Western cultural upbringing, limited education

Dementia and MCI are **clinical** diagnoses. While the tests were validated against a cut-off score, in real life the score should be used as a guidance and is not diagnostic.

We get asked this every year...

- **What is better to test for executive function? MoCA, MMSE, or Clock draw test (CDT)?**
 - As always, read the wording of the prompt carefully.
 - The more specific test for executive function is the clock draw:
 - you can score quite poorly on the MoCA due to impairments in other domains (e.g. memory, language, etc.) while your executive function remains preserved.
 - The more sensitive test for executive function is the MoCA, given it has multiple tasks that assess for it (e.g. mini Trails B, CDT, abstraction, letter fluency).
 - Thus, the “better” test overall depends on whether you want to prioritize specificity or sensitivity.

Workup for dementia

History:

- Ask about domains, collateral, functional history, **screen for safety (see management)**
- **Reversible causes/confounders: mood, sleep (esp undiagnosed sleep apnea), meds, alcohol**
- Informant/collateral history – vital! Can use validated tool (eg) Quick Dementia Rating System

Physical examination:

- Cognitive testing, mental status examination
- Neurologic examination, with emphasis on UMN findings and parkinsonism

Investigations: Rule out confounders/ reversible causes

- CBC, lytes, Cr, calcium/albumin, LFTs, blood glucose/ A1c, TSH, vitamin B12
- +/- HIV, VDRL
- Sleep history → Sleep study, hearing assessment, vision assessment

Imaging: Not required for all - see next slide

- MRI preferred to CT if suspected vascular dementia, and in some subtypes of dementia and other more rare conditions

Biomarkers/Genetic testing: Limited role currently, will change with novel therapies in the future

When to image: “BrAIN”

Bleeding Risk

- Head trauma
- Anticoagulant use (or bleeding disorder)

Abnormal Presentation

- Age <60 years
- Rapid, unexplained decline (1-2 months)
- Shorter duration of dementia (<2 years)
- Unusual or atypical cognitive presentation

Intracranial Lesion

- History of cancer
- Unexplained focal neuro signs or symptoms (headache, seizure)

Normal Pressure Hydrocephalus (NPH)

- Gait disturbance, incontinence

Practically, almost all get imaging unless classic picture of Alzheimer’s Dementia .

MRI recommend over CT for evaluation of microangiopathic changes. (CCCD 2020 Guidelines)

PLUS if the presence of cerebrovascular disease would change clinical management

Dementia Syndromes



Alzheimer's Dementia (AD)

Abnormal protein: β -amyloid

Diagnosis:

- Dementia AND
- **Insidious onset, gradual progression** (years)
- Initial and most prominent deficits in:
 - **Amnestic AD** (starts with short-term memory deficit, most common)
 - Non-amnestic AD (do not memorize)
 - Logopenic Primary Progressive Aphasia (language deficit, trouble with naming and sentence repetition)
 - Frontal variant AD (fronto-executive deficit)
 - Posterior Cortical Atrophy (visuospatial deficit)

BONUS

FYI for clinic, less likely on exam

Biological Definition of AD:

ATN (Amyloid, Tau, Neurodegeneration) Classification

Three groups of biomarkers (by CSF, PET, or serum)

1. Amyloid
2. Tau
3. Neurodegeneration

A+ T- (N)-	Alzheimer's pathologic change with dementia
A+ T+ (N)-	Alzheimer's disease with dementia
A+ T+ (N)+	
A+ T (N)+	Alzheimer's and concomitant suspected non Alzheimer's pathologic change with dementia
A- T+ (N)-	non-Alzheimer's pathologic change with dementia
A- T- (N)+	
A-T+ (N)+	

Why is this important?

- **Blood-based biomarkers (BBM)** becoming more accessible
- New practice guidelines from Alzheimer Association (**July 2025**): for patients with objective cognitive impairment presenting to **specialized memory care**...
 - suggests using a high-sensitivity BBM test as a triaging test in workup of AD.
 - suggests using a high-sens and high-spec BBM test as a confirmatory test in workup of AD.
 - A BBM test should not be obtained before a comprehensive clinical evaluation by a health care professional, and test results should always be interpreted within the clinical context.
 - Practical pearl: still +-variability in accuracy of commercially available BBM tests!
 - **Should be ordered by Dementia specialist, not in GIM scope given wording of recommendation = "Refer to Dementia Specialist" in oral scenario on your exam**

Jack et al. Alz and Dem. 2018.

Jack et al. Alz and Dem. 2024.

Alzheimer's Association International Conference. 2025.

Alzheimer's Dement. 2025;21:e70535.

Vascular Dementia

Diagnosis:

- Cognitive impairment (any domain, but often **fronto-executive**) AND
- **Imaging** evidence of cerebrovascular disease
 - Strokes
 - White matter changes
 - Microhemorrhages (typically subcortical) may be seen with severe uncontrolled hypertension
- +/- Temporal relationship

Two main syndromes:

- 1. Post-stroke vascular dementia** – “step-wise decline”
- 2. Subcortical ischemic syndrome** – more common, insidious onset
 - *Imaging*: periventricular white matter changes “moderate micro-angiopathic changes” +/- lacunar infarcts
 - *Clinically*: frontal-dysexecutive syndrome
 - *Supportive findings*: insidious onset, gait disturbance, “**slow**”

Dementia with Lewy Bodies (DLB)

Abnormal protein: α -synuclein

Dementia and 2 of:

- Fluctuating cognition (attention, alertness)
- Recurrent, well-formed visual hallucinations
- Parkinsonism (bradykinesia, rest tremor, rigidity)
- REM sleep behaviour disorder
 - Acting out their dreams in sleep, often vivid, shouting or striking out (normally in REM sleep body is “paralyzed” and atonic)
 - Can precede dementia by years or decades

“1 year rule” to distinguish DLB and Parkinson’s disease dementia (PDD): if dementia precedes or begins within 1 yr of onset of parkinsonism → DLB

May also have 1 clinical feature and 1 biomarker, including:

- Low dopamine uptake in basal ganglia (SPECT or PET scan)
- Polysomnographic confirmation of REM sleep without atonia
- Cardiac sympathetic denervation on MIBG myocardial scintigraphy

Supportive but non-diagnostic features:

Sensitivity to antipsychotics, postural instability, repeated falls, severe autonomic dysfunction (constipation, urinary incontinence, orthostasis, syncope), hyposmia, hallucinations or delusions, apathy, anxiety/depression

Frontotemporal Dementia

Abnormal protein: Tau most common, TDP-43

Diagnosis of Behavioural Variant FTD (bvFTD):

– Dementia and 3 of:

- Disinhibition
- Apathy
- Loss of empathy
- Perseveration (Perseverative behaviours – may be unable to switch appropriately between tasks)
- Hyperorality (Putting things into mouth, or eating indiscriminately)
- Executive dysfunction

Note there are three subtypes of FTDs:

1. bvFTD
2. Semantic Variant Primary Progressive Aphasia (svPPA): left anterior inferior temporal lobe atrophy → loss of conceptual knowledge
3. Non-fluent Primary Progressive Aphasia (nfPPA): left fronto-insular atrophy → impaired speech production

Mixed Dementia

Diagnosis:

- Dementia
- **Any combination** of syndromes
- Most common:
 - Alzheimer's clinical syndrome AND
 - Imaging findings consistent with vascular dementia

MCQ #1 2026

An 81-year-old female is evaluated for gradually worsening memory and functional decline over the past 3 years. Her daughter reports that she struggles with misplacing her wallet, paying bills and planning meals. She becomes irritable when the daughter tries to intervene and this has led to several verbal arguments between them. Her medical history is notable for HTN, T2DM, DLD, peripheral neuropathy, and GERD. Medications include metformin, empagliflozin, rosuvastatin, ramipril, pantoprazole. She sometimes appears slowed in thought and movement, and his family has noticed that her walking speed is reduced. On cognitive testing, she has difficulty with clock drawing and mini Trails B. Which of the following is the most likely diagnosis?

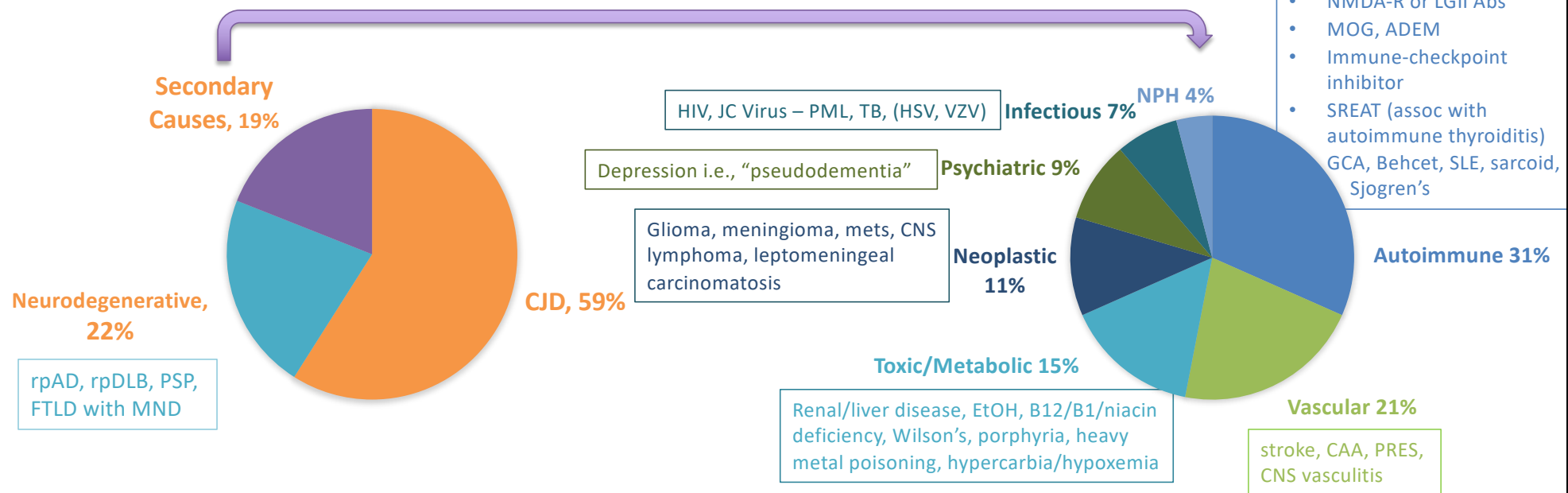
- A. Major depressive disorder
- B. Frontotemporal dementia
- C. Vascular dementia**
- D. Alzheimer disease

The correct answer is C. This patient's presentation is most consistent with subcortical vascular dementia which is gradual and not step-wise. The key clues are her vascular risk factors (HTN, diabetes, dyslipidemia), progressive executive dysfunction (difficulty with attention, planning, bills, clock drawing, Trails B), slowed cognitive processing, and gait changes. Irritability is situational and not indicative of FTD. Don't be fooled by "gradually worsening memory" always being AD – on exam and in practice – it may be an attention issue.

Rapidly Progressive Dementia

Evolving “more rapidly than expected for dementia syndrome”

- **Weeks to 1-2 years** from first symptom or functional change to dementia



Day and Tang-Wai. Neurodegener Dis Manag. 2014.

Uy et al. Pract Neurol. 2021.

Day. Continuum. 2022.

Work-up of Rapidly Progressive Dementia

- History (med review!) and Physical Exam
- **EVERYONE:**
 - CBC, lytes, ext lytes, glucose Cr BUN, LFTs, TSH, CRP, B12, HIV, VDRL, blood/urine tox
 - MRI brain (include T1, T2, FLAIR, DWI, GAD)
 - LP (cells, protein, glucose, C&S, VZV, HSV, oligoclonal bands, cytology)
 - EEG
- Consult Neurology
- If above negative and suspect:
 - *CJD*: CSF RT-QuIC, 14-3-3 protein
 - *Neurodegenerative process*: CSF tau, phosphorylated tau, amyloid- β 42, amyloid PET
 - *Paraneoplastic syndrome*: CT CAP, testicular US, paraneoplastic panel (CSF AND serum!!), whole-body PET-CT
 - *Vasculitis*: MRV/A
 - *Heavy metal poisoning*: ceruloplasmin, ferritin, iron studies, urine heavy metals
 - *Infectious*: CSF and serum Lyme, cryptococcal Ab, T. whipellii PCR, blood cultures
 - *Autoimmune*: ANA, ANCA, SPEP, anti-TPO, anti Thyroglobulin
- If still don't know and patient rapidly decompensating + identifiable focus of abnormality on MRI: brain or leptomeningeal biopsy. Time is brain!

?CJD – ordering 14-3-3 τ protein in CSF requires ++ lab notice and permission from Health Canada



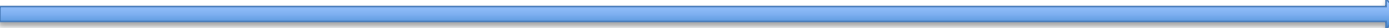
BONUS

Read on own

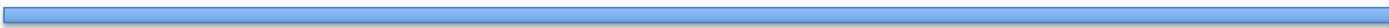
Rapidly Progressive Dementia: Spotlights

Both of these entities are rare – 1-2/million per year

Spotlight on Creutzfeldt-Jakob Disease CJD^{1,2,3}

- Prion-related disorder – 85% sporadic, 13% familial, 1-2% infectious/iatrogenic. Median age onset 60s.
- Rapidly progressive (weeks to months) and uniformly fatal, median survival 5 months
- Clinical:  **Death**
Sleep/wake disturbance, dizziness/vertigo, depression/anxiety, personality change Visual disturbances: blurred vision, diplopia, hallucinations, **metamorphosis** Dementia, (startle) **myoclonus**, **akinetic mutism**
Gait/balance disturbance, aphasia, ataxia, nystagmus
- MRI typical findings (83-92% sens, 97-95% spec): “hockey stick” sign – hyperintensities in thalamic pulvinar nuclei and dorsomedial thalami, “**cortical ribboning sign**” - curvilinear hyperintense signal involving the [cerebral cortex](#) on [DWI](#) images with a corresponding low signal on [ADC](#) images.
- Diagnostic: CSF markers of rapid neuronal degeneration (most commonly **14-3-3 tau protein**, there are others beyond scope of GIM), periodic sharp wave complexes on EEG (64% sens, 91% spec)

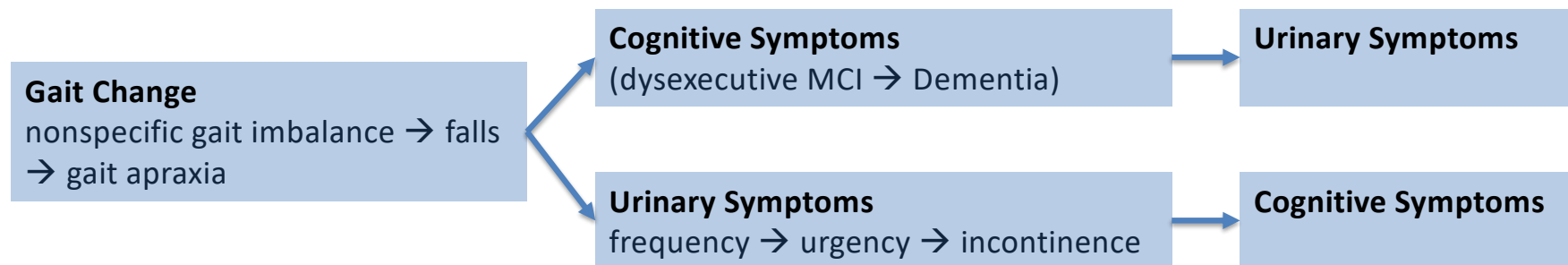
Spotlight on Ant-NMDA Receptor Encephalitis (“Brain on Fire”)^{4, 5}

- 60% associated with ovarian teratomas, rare cases are non-para neoplastic. Median age onset 20s.
- Clinical:  **85% survival with immune therapy**
Hypersomnia, **mania**, bizarre, disinhibited behaviour, aggression, pressured speech, disorganized thinking, catatonia **Seizures**, Facial dyskinesias, dysautonomia, hypoventilation Prolonged: executive dysfunction, impulsivity, disinhibition

Normal Pressure Hydrocephalus

- “Classic” Triad is actually a late presentation: Gait apraxia (magnetic gait), executive dysfunction (dementia), and urinary incontinence

Typical Course:



- Imaging: Ventriculomegaly (must be differentiated from ex vacuo) + other features
- No recent consensus diagnostic criteria; response to large volume lumbar puncture (gait, less so cognition) helps to determine if VP shunt indicated for permanent management

Ib. Dementia and Mild Cognitive Impairment: Management



Key Guidelines:

1. *Lancet* Commission report on Dementia prevention, intervention, and care (2024)
2. WHO Guidelines on Risk reduction of cognitive decline and dementia (2019)
3. Recommendations of the 5th Canadian Consensus Conference on the diagnosis and treatment of dementia (2020)
4. Canadian Stroke Best Practice Recommendations: Vascular cognitive impairment guidelines update (2025)
5. Canadian Clinical Practice Guidelines for Assessing and Managing Behavioural and Psychological Symptoms of Dementia (2024)
6. CMA Driver's Guide 10th Edition, Section 8 (2023)

MCQ #2 2026

A 76-year-old male is evaluated in clinic. He received a diagnosis of Parkinson's Disease 7 years ago with onset of left hand tremor. He was started on levodopa-carbidopa, which was last increased to levodopa-carbidopa 100/25mg 2 tablets QID one year ago with good motor response. Over the past year, his family has noticed worsening forgetfulness, difficulty with planning, and occasional visual hallucinations which are not bothersome. He also reports worsening nausea. He remains independent with basic activities of daily living but now struggles with managing his groceries, medications and finances. Neurologic examination shows masked facies, left hand tremor, and a shuffling gait. Cognitive testing reveals deficits in executive function, attention and delayed recall.

Which of the following is the most appropriate next step?

- A. Decrease levodopa-carbidopa 100/25mg to 2 tablets TID
- B. Start quetiapine 25mg po QHS
- C. Start metoclopramide 10mg po TID
- D. Start donepezil 2.5mg po daily



Non-pharmacologic management

All Recommended³ for general health benefit, esp. given minimal harms and significant heterogeneity in how they were studied.

	MCI	Dementia	Comments
Exercise	SRMA evidence for reducing risk of cog decline; recommended in guidelines ^{1,3,4,5}	Not per guidelines, mixed SR evidence ^{6,7,8}	Some improvement in ADLs, BPSD in dementia possible ^{2,6} in addition to numerous well-recognized health benefits
Cognitive Training	Evidence from small RCTs, mixed from SR ^{1,3,9,10}	Small, short-term benefit in mild-mod dementia ^{2,3,11,12}	Generally improves scores in trained domain, but clinical relevance is questioned
Social Engagement	Insufficient evidence ¹	Insufficient evidence ¹³	Very small effect sizes, but guidelines recommend for general good health and wellbeing throughout life-course
Mediterranean Diet	May reduce decline per WHO ¹ but not Lancet Commission ²	No evidence	
Hearing	Mixed guidelines ^{1,2,3} , ↓ conversion to dementia in systematic rev ¹⁴	No evidence, but may improve QoL, depression ^{14,15}	Generally recommended to screen and treat widely. Correct hearing with hearing aids or cochlear implants.
CPAP for OSA	Low quality evidence of some benefit ¹⁶	No evidence	

1. WHO Guidelines. 2019
2. Lancet Commission. 2024
3. 5th CCCDTD. 2020.
4. Abbass Tahami Monfared Neur & Ther 2023
5. Li et al. Clin Psychol Rev. 2025
6. Lamb et al. BMJ. 2018.
7. Forbes et al. Cochrane. 2015.
8. Sanders et al. PLoS One. 2019
9. Butler et al. Ann Int Med. 2018.
10. Belleville et al. JAGS. 2018.
11. Bahar-Fuchs et al. Cochrane. 2013.
12. Cafferata et al. Psychol Bull. 2021
13. Woods et al. Cochrane. 2023
14. Sheng et al. JAMA Neurology. 2023
15. Atef et al. J Alzheimers Dis. 2023
16. Seda et al Curr Neuro & NeuroSci Reports, 2021

Pharmacologic management

	MCI	Dementia	Comments
Over the counter supplements	No ^{1,2}	No ¹	Negative studies for: vitamin B, folate, vitamin E, ginkgo biloba, flavonoids, omega-3
Cardiovascular medications	No evidence for statins, ASA, or glycemic control² <i>Exception: BP control <140 in vascular dementia, ASA for vascular cognitive impairment with prior stroke (not just microvascular changes)^{3,5}</i>	No evidence.	SPRINT-MIND ⁴ (excluded stroke, dementia, and diabetes; target SBP <120) showed decreased MCI and combined MCI/probable dementia, but no significant decrease in dementia.
Cognitive Enhancers (ChEI, memantine)	No ³	Yes for specific dementia syndromes ^{3,5}	See next slides

1. WHO Guidelines. 2019
2. Lancet Commission. 2024
3. 5th CCCDTD. 2020.
4. Williamson et al. JAMA. 2019.
5. 5th CCCDTD Vascular Cognitive Impairment Guidelines. 2020.

Pharmacotherapy for Alzheimer's Dementia



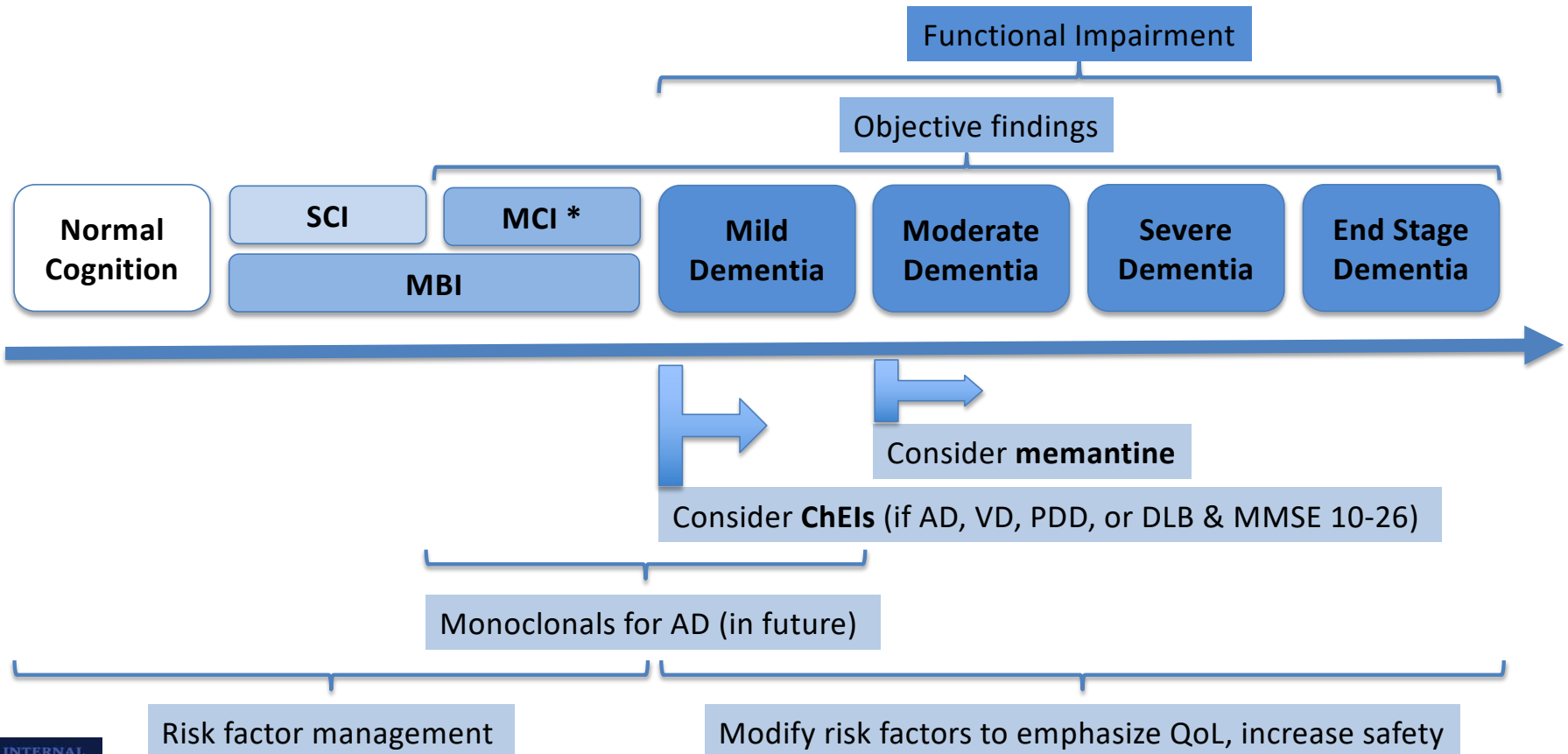
- **Cholinesterase Inhibitors** (Donepezil, Galantamine, Rivastigmine)
 - Health-Canada approved for “mild, moderate and severe dementia”. In practice, a trial is recommended for most patients with AD and MMSE 10-26.^{1,2}
 - Modest improvements in cognition, function, global assessment & behaviour, unplanned time in hospital¹
 - **Side Effects:** GI intolerance (anorexia), weight loss, urinary incontinence, wild or vivid dreams, bradycardia
 - Avoid if: conduction defects (except RBBB), bradycardia, or unexplained syncope. Use with caution in asthma, COPD, seizures, GI bleeds, urinary incontinence
 - **When to discontinue^{2,3*}:** Intolerable side effects, lack of benefit, severe or end-stage dementia, cognitive decline over 6 months not explained by other precipitant
 - How to discontinue: taper by 50% every 4 weeks and monitor for clinical worsening
 - NB: Discontinuation of ChEI in pts not experiencing side effects is associated with worse short-term cognition, worse function at one year, and increase in BPSD in short- and medium-term.⁴ New longitudinal studies indicate benefit of cholinesterase inhibitors even in severe dementia.¹ NICE 2018 also says shouldn't stop cholinesterase inhibitors based on disease severity alone but follow Canadian guidelines for exam.

Pharmacotherapy for Alzheimer's Dementia

- **NMDA-Receptor Antagonist** (Memantine)
 - Health Canada-approved for moderate to severe AD
 - Cochrane review: In mod-severe AD (defined as MMSE 5-13), small improvement in cognition, less agitation, no difference in function¹
 - Mixed evidence re: **additive** benefit with ChEIs.^{1,2,3}
 - Older Canadian guidelines say addition to donepezil is “rational, **but not enough evidence** to recommend for or against.”⁴ Updated guidelines don't comment.⁵
 - **Side Effects:** Dizziness (6.1% vs 3.9%), Headache (5.5% vs. 4.3%), no diff in falls¹
 - NB: Costs \$3-5 per day. If no private drug coverage, often encourage family to use money on other services instead (e.g. day programs, transportation, PSW).

1. Cochrane Database Syst Rev. 2019 Mar;3(3):CD003154. 2. Brain Behav. 2020 Nov;10(11):e01831. 3. DOMINO-AD. N Engl J Med. 2009 Jul 24;10:57. 4. Can Fam Physician. 2014 May;60(5). 5. 5th CCCDTD. Alzheimers Dement. 2020 Nov;11;6(1):e12056.

Pharmacotherapy: WHEN?



BONUS

Read on own

A note about monoclonal antibodies: Aducanumab, Lecanumab, Donanemab

- mAbs target amyloid- β (protofibrils or aggregates/plaques)
- Indicated/studied for **amyloid-positive amnestic MCI or mild AD: lecanemab** (CLARITY-AD), **donanemab** (TRAILBLAZER-ALZ2), and the earlier **aducanumab** (EMERGE/ENGAGE).
 - Efficacy: modest slowing of clinical decline in phase-3 trials; clinical meaningfulness and duration of benefit remain under evaluation.
 - Safety: Amyloid-related imaging abnormalities - ARIA (edema and hemorrhage) is the major risk — **lecanemab ARIA-E $\approx$ 13% (any ARIA $\approx$ 21%), donanemab ARIA-E $\approx$ 20–24%** (radiographic rates; symptomatic ARIA rates are lower).



Canada: aducanumab was not adopted in Canada; **lecanemab was just approved by Health Canada in November 2025; donanemab is still under Health Canada review. Thus far** access in Canada has mainly been via RCTs, special access/compassionate programs, or is limited while regulatory and reimbursement decisions are made.

- FYI: Lecanemab approved in Europe, UK, Japan and China; Donanemab approved in UK and Japan
- Systems issues: require access to biomarker testing upfront (MRI amyloid-PET), infusion centre, drug cost (>\$25,000 /yr)

Pharmacotherapy for Vascular Dementia

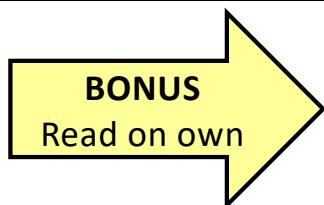
- Optimization of **vascular risk factors** (nonpharm & pharm)
- **ChEIs (& memantine)**
 - 2020 Canadian Consensus Recommendations suggest that ChEIs and memantine “may be considered” given diagnostic uncertainty and lack of other options
 - Indicated in **mixed AD & vascular dementia**
- ASA is not indicated with white matter changes only, “reasonable to use” if lacunar infarcts detected but uncertain benefit

Pharmacotherapy for Dementia with Lewy Bodies (DLB) & Parkinson Disease Dementia (PDD)

- **ChEIs**
 - Good evidence cognition, global assessment, and behavioural disturbance, **hallucinations**
 - 1st trials used rivastigmine, but **donepezil** is better tolerated
- **Levodopa/carbidopa**
 - Greater benefit for movement in PDD than DLB
- **Memantine: Can be considered second line¹**
 - Consensus, not good evidence

Pharmacotherapy for Behavioural Variant Frontotemporal Dementia (bvFTD)

- **NO medication has been shown to help the cognitive decline in FTD**
 - ChEIs contraindicated/can make worse
- **Trazodone or SSRIs** for behaviours (cautiously)
- Often paradoxical worsening to antipsychotics
- Androgen Deprivation Therapy for sexual disinhibition



Summary of Pharmacologic Treatment



Dementia	Pharmacotherapy
Alzheimer's Dementia	ChEI indicated
Vascular Dementia	Optimize vascular risk factors Consider ChEI, indicated in mixed AD + VaD
Dementia with Lewy Bodies	ChEI indicated for cognition, hallucinations & behaviours
Parkinson's Disease Dementia	ChEI indicated , similar evidence as DLB
Frontotemporal Dementia	Trazodone, SSRIs (for behaviours only) No role for ChEI

MCQ #2 2026

A 76-year-old male is evaluated in clinic. He received a diagnosis of Parkinson's Disease 7 years ago with onset of left hand tremor. He was started on levodopa-carbidopa, which was last increased to levodopa-carbidopa 100/25mg 2 tablets QID one year ago with good motor response. Over the past year, his family has noticed worsening forgetfulness, difficulty with planning, and occasional visual hallucinations which are not bothersome. He also reports worsening nausea. He remains independent with basic activities of daily living but now struggles with managing his groceries, medications and finances. Neurologic examination shows masked facies, left hand tremor, and a shuffling gait. Cognitive testing reveals deficits in executive function, attention and delayed recall.

Which of the following is the most appropriate next step?

- A. Decrease levodopa-carbidopa 100/25mg to 2 tablets TID
- B. Start quetiapine 25mg po QHS
- C. Start metoclopramide 10mg po TID
- D. Start donepezil 2.5mg po daily**

The correct answer is D. This patient with long-standing Parkinson's disease now has Parkinson's disease dementia. The presence of mild, non-distressing hallucinations does not require antipsychotic therapy at this time. Levodopa-carbidopa should not be reduced given good motor response and ongoing motor symptoms. Metoclopramide is contraindicated in Parkinson's disease as a dopamine antagonist. Cholinesterase inhibitor can improve cognition and function in PDD, donepezil is best tolerated re GI side effects.

Complications of Dementia

- Behavioral and psychological symptoms of dementia: [see next slides]
- Swallowing and Weight loss:
 - Dietician for weight loss. Consider nutritional supplement (ex. Ensure) between meals.
 - Refer to SLP for suggestions and feeding techniques
 - **Enteral Tube feeding¹**: **No evidence** of improved survival, comfort, BPSD, QoL or caregiver outcomes. Increased risk pressure ulcer, one study found increased risk mortality.
- Safety concerns:
 - Environmental modifications (stove, faucets, falls; via OT home safety referral)
 - Wandering (Medic-Alert bracelets, geo-locating devices, wander registry)
 - Weapons/firearms and power tools
 - Finances (scams, POA for finances, tax credits; via SW referral)
 - Medications (blister packs, assisted/supervised administration; via RN/PharmD)
 - Driving [see next slides]
- Caregiver burnout: support services (referral to other local services)

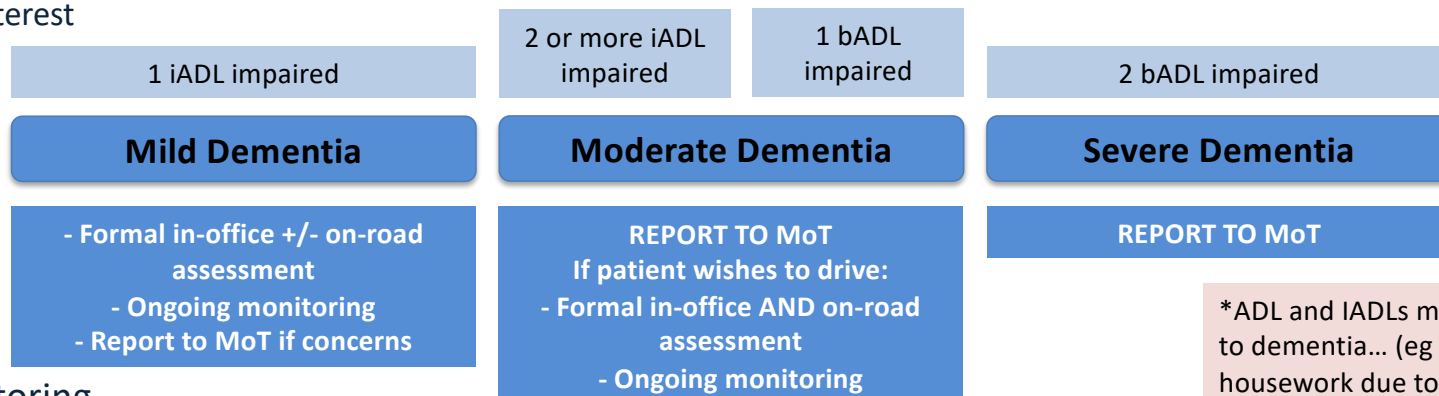
Davies et al. Cochrane. 2021

Driving in Dementia

(MoT = Ministry of Transportation in your jurisdiction)

CMA Driver's Guide 10th Edition

- Understand requirements in your jurisdiction
- Clinical decision-making: <https://www.drivinganddementia.ca/assessingfitnesstodrive>
 - Diagnosis of dementia alone is not sufficient to withdraw driving privileges
 - No in-office test or battery of tests, incl. MMSE or MoCA, has sufficient sensitivity or specificity to be used as a sole determinant of driving ability, but may indicate need of further assessment
 - NB. Caregiver concern = take seriously; No caregiver concern = doesn't hold much water, they may have conflict of interest



*ADL and IADLs must be impaired due to dementia... (eg if not doing heavy housework due to arthritis this doesn't track for multiple iADLs)

- Monitoring
 - Re-evaluate if deemed fit to drive every 6 to 12 months or sooner
 - Driving cessation associated with social isolation, depression, and other adverse health outcomes; monitor for these

Future Planning

- Discuss natural history of disease
- Goals of care and advanced directives (eg, code status, POA for personal care)
 - Advancecareplanningontario.ca has good resources for these discussions
- Other advance care planning (eg, completing a will, feeding tubes, institutionalization, MAiD)

Waiver of Final Consent

Legal with passing of Bill C-14 March, 2021; Track 1

Someone has already been assessed and deemed eligible for MAiD, can request to receive MAiD on their chosen date should they become incapacitated for consent at the time of their procedure.

Track 1:

- The person has a grievous and irremediable medical condition
- The person's natural death is reasonably foreseeable

"Advance Request for MAiD"

Not legal in Canada (other than QC, Bill 11 as of Oct 30, 2024)

e.g. Someone living with dementia who is currently capable states "If I become incapable and am dependent for toileting or do not know names of my spouse and children, I would want MAiD".

MCQ #3 2026

A 77-year-old woman with moderate Alzheimer dementia is brought to clinic by her son because of escalating behavioral symptoms. Over the past several months, she has become increasingly irritable, verbally agitated, and resistant to care. These behaviors occur weekly, causing moderate distress to her caregiver. She does not have hallucinations or delusions. Structured activities, caregiver support, environmental modifications have been tried for several months with limited benefit. She has no history of arrhythmia or QTc prolongation.

Which of the following is the most appropriate pharmacologic intervention at this time?

- A. Quetiapine 25mg po daily
- B. Aripiprazole 2mg po daily
- C. Haloperidol 1mg po daily
- D. Citalopram 10mg po daily
- E. Trazodone 25mg po daily

Behavioural & Psychological Symptoms of Dementia (BPSD)

- Clusters of symptoms:
 1. **Psychosis:** delusions, hallucinations, suspiciousness
 2. **Aggression:** verbal, physical, defensive, resistance to care
 3. **Agitation:** restlessness, anxiety, vocalization, repetitive actions, pacing, wandering, hoarding
 4. **Depression:** sadness, guilt, hopelessness, irritability, suicidality
 5. **Mania:** irritability, euphoria, pressured speech, sexual disinhibition, impulsiveness
 6. **Apathy:** amotivation, withdrawn

Behavioural & Psychological Symptoms of Dementia (BPSD)

- Every behaviour is the communication of an **unmet need**
- Non-pharmacologic:
 - PIECES approach: identify contributing Physical, Intellectual, Emotional, Capabilities, Environmental, and Social factors¹
 - Recent network meta-analysis demonstrated non-medication treatments (e.g. outdoor therapy, music and massage, massage and touch) were *as or more efficacious* than medications

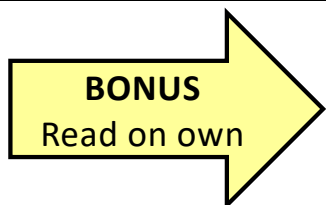
Behavioural & Psychological Symptoms of Dementia (BPSD)

*2024 Guidelines focus on Alzheimer's Disease and related dementias

Symptom	Non-pharm	Pharm – Effective trial = 8 weeks, at least 2 weeks at therapeutic dose.
Agitation	✓ Interdisciplinary approach ✓ Music therapy ✓ Massage therapy ✓ Robotic pets ✓ Animal therapy ✓ Physical exercise ✓ Aromatherapy ✗ Seclusion ✗ Physical restraint	Moderate: ✓ Citalopram (≤20 mg/d) (not escitalopram) Severe: ✓ Aripiprazole, brexpiprazole (max 3mg/d)*, risperidone (max 1 mg BID)* ● Quetiapine if above not tolerated due to EPS ● Haloperidol (2-6 mg/d) if refractory to above ● Nabilone if refractory to above ● Carbamazepine if refractory to above ✗ Olanzapine (other than short-term/emergency) ✗ ChEI, memantine ✗ Trazodone, mirtazapine, fluoxetine, paroxetine, TCAs ✗ Valproic acid ✗ Depot antipsychotics *Approved by Health Canada if 3/3 of: 1. pure AZD, 2. non-pharm Rx ineffective 3. risk to self/others or distressing psychotic symptoms
Psychosis		Moderate: ✓ Citalopram (≤20 mg/d) Severe: ● Aripiprazole or risperidone*
Depressive symptoms	✓ Interdisciplinary approach ✓ Robotic pets ✓ Animal therapy ✓ Physical exercise ✓ Aromatherapy ✓ Massage therapy ✓ Cognitive stimulation therapy ✓ Reminiscence therapy ✓ OT involvement ✓ Home-based problem-based therapy	✓ Moderate-to-severe: antidepressants (i.e., SSRIs) if no benefit from non-pharma ✗ Pharma Tx depressive symptoms in dementia without concurrent Dx depression

BLACK BOX WARNING for ALL ANTI-PSYCHOTICS:
Increased stroke risk 2.7x, Increased risk of death 1.6x, NNH (stroke, death) = 100
2x risk pneumonia, 1.7x risk AKI, 1.62x risk VTE, 1.43x risk fracture, 1.3x risk MI²

1. CCSHM BPSD Guidelines, 2024. 2. BMJ 2024; Apr 17:385:e07626



Behavioural & Psychological Symptoms of Dementia (BPSD)

Symptom	Non-pharm	Pharm – Effective trial = 8 weeks, at least 2 weeks at therapeutic dose.
Anxiety	☒ CBT ☒ Music therapy	☒ Moderate to severe: Citalopram (≤ 20 mg/d)
Sexual expressions of potential risk	☒ Caregiver education, removal of environmental triggers	☐ Neither for nor against pharmacologic interventions.

- Consider taper/deprescribing antipsychotics after 90 days because nature/intensity of BPSD tend to change over time.
- Decrease dose by **25-50% every 1-2 weeks** until either discontinued or BPSD worsen (in which case return to minimal effective dose).

MCQ #3 2026

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- A. Quetiapine 25mg po daily
- B. Aripiprazole 2mg po daily
- C. Haloperidol 1mg po daily
- D. Citalopram 10mg po daily**
- E. Trazodone 25mg po daily

The correct answer is D. This patient has moderate BPSD characterized by agitation without psychosis. According to the 2024 CCSMH BPSD guidelines, citalopram is recommended for both agitation and psychosis of moderate severity, and in severe agitation where risks of antipsychotics preclude their use. Start at 10mg and uptitrate to 20mg.

II. Delirium

“Acute brain failure”



Key Guidelines:

1. Canadian Coalition for Seniors Mental Health (CCSMH) Assessment and Treatment of Delirium (2014 update).
2. American Geriatrics Society Clinical Practice Guideline for Postoperative Delirium in Older Adults (2014).
3. NICE Clinical Practice Guideline Delirium: prevention, diagnosis and management. (2010, **2023 update**).

Delirium: Diagnosis

Core features (DSM-V):

- A. **Disturbance in attention** (reduced ability to direct, focus, sustain, and shift attention) and awareness (reduced orientation to the environment).
- B. Develops over a **short period of time** (usually hours to a few days), represents a change from baseline, and **tends to fluctuate** in severity during the course of a day.
- C. Additional disturbance in cognition (memory deficit, disorientation, language, visuospatial ability, or perception).
- D. Disturbances in Criteria A and C are not better explained by another preexisting, established, or evolving neurocognitive disorder and do not occur in the context of a severely reduced level of arousal, such as coma.
- E. Hx, P/E, or labs suggest disturbance is a direct physiological consequence of another medical condition, substance intoxication or withdrawal, or exposure to a toxin.

Delirium Screening



CAM¹ **1 and 2 + either 3 or 4**

1) Inattention

(examination: months backwards)

2) Acute onset and fluctuating course

(history: collateral RN/family)

3) Disorganized thinking

(observation: Patient making sense? “can a stone float on water?”)

4) LOC altered

(observation: hypervigilant? somnolent? both?)

Memorize!
mnemonic “IADL”

Pros/Cons of CAM: need collateral re: acuity. Can't score if drowsy.
Validated for use with interpreter, and known baseline dementia.

4AT² **Score 1-3: cognitive impairment** **Score 4+: delirium +/- cog impair't**

1) Alertness

0: Normal or mild sleepiness lasting <10sec after waking then normal

4: Clearly abnormal

2) Age, DOB, name of hospital/building, year

0: No mistakes

1: 1 mistake

2: 2+ mistakes or untestable

3) Attention:

0: Month of Year Backward (MOYB) to July

1: MOYB <July or refuses to test

2: Untestable (drowsy, inattentive or too unwell)

4: Acute change or fluctuating course:

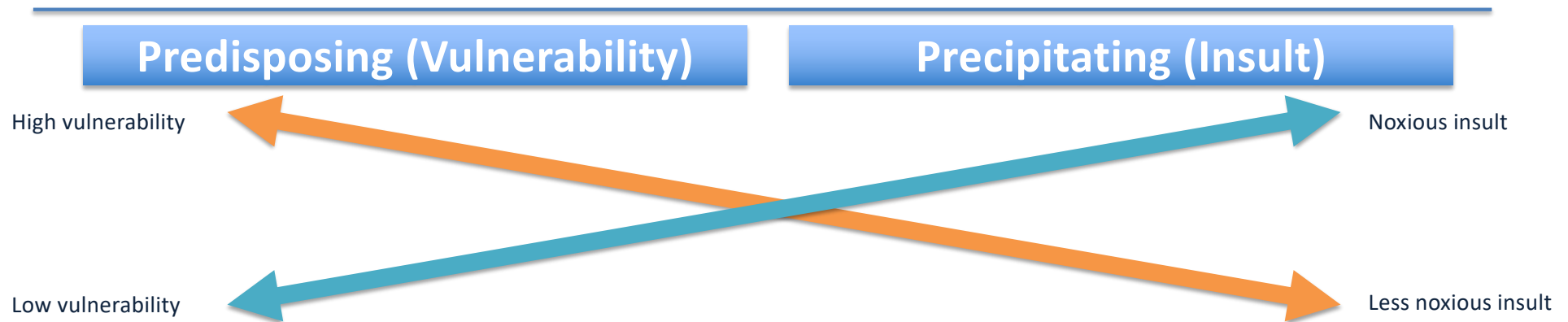
(within last 2 weeks and persisting in last 24h)

0: no

4: yes

1. Ann Int Med. 1990. Dec; 113 (12): 941-948
2. Age Aging. 2021. May; 50(3): 733-743.
3. Geriatr Gerontol Int. 2016 Sep;16(9):1028-35.

Delirium: Common Risk Factors



Predisposing:

- Age >70 (RR 4.0)
- Known cognitive impairment (RR 2.8)
- Functional impairment (RR 4.0)
- EtOH misuse (RR 5.7)
- Sensory impairment (RR 1.1-3.5)

Precipitating:

- Drugs (RR 2.9-4.5)
- Physical restraints (RR 4.4)
- Bladder catheter (RR 2.4)
- Infection (RR 3.1)

Review prior to
oral exam

Delirium: Prevention / Non-Pharm

Targeted Risk Factor	Strategies
Cognitive impairment	Orientation protocols, provision of clocks and calendars, family visits, pre-op cognitive training
Sensory impairment	Provide glasses, hearing aids
Malnutrition	Dentures, assistance with feeding, positioning
Fluid/Electrolyte imbalances	Volume assessment; normal glucose, Na, K, Ca
Agitation	Provide redirection, correct over/under-stimulation (move to quiet room, provide tasks such as folding towels), family visits
Sleep deprivation	Inpatient Unit strategies to reduce noise; Schedule medications and procedures to allow for proper sleep; Non-pharmacologic measures to prevent sleep deprivation
Functional impairment	Mobilize early! PT, OT
Pain	Scheduled analgesia; judicious use of opioids
Iatrogenic complications	Remove urinary catheters, screen for retention/incontinence; Skin care regimen; Bowel regimen; Chest physiotherapy; Treat nosocomial infections; EEG-guided anesthesia ⁴ ; pre-op cognitive rehab ⁵ Note: <u>No difference</u> in delirium with regional vs general anesthesia! ⁶
High-risk medications	Avoid/ Minimize benzodiazepines, anticholinergics, antihistamines

Delirium: Work-up

Assess for cause(s) and contributors

Indications for Neuroimaging

- 1) Focal neurological deficits
- 2) Reduced level of consciousness
- 3) Fall or head trauma
- 4) Anticoagulation
- 5) Fever + acute change (If CNS infection is suspected)
- 6) No other cause identified

Scottish Intercollegiate Guidelines Network 2019 Delirium Guidelines

Etiology	History, Physical Exam/Investigations
<u>D</u>rugs	Too much: opioids, benzodiazepines, sedatives, anticholinergics, intoxications/recreational, Too Little: Pain undertreated, withdrawal (EtOH, benzo, opioid, nicotine) <i>+/- ASA/Tylenol/EtOH levels, tox screen</i>
<u>I</u>nfection/ Inflammation	<i>Vitals, focal signs/sources of infection, cultures as appropriate (blood, urine, Chest XR, +/- LP)</i>
<u>M</u>etabolic	<i>CBC, lytes, Cr, LEs/LFT, Calcium, VBG, glucose, TSH, B12, cortisol (hypoactive)</i>
<u>E</u>nvironment	Sensory deprivation (windowless room, hearing/vision aids), sensory overload, isolation from familiar surroundings, absence of orientation (clock/calendar), restraints, disruption to rest/sleep
<u>R</u>etention	<i>Abdominal XR, bladder scan</i>
<u>S</u>tructural abnormality	Neuro: seizure, stroke, hemorrhage (<i>exam, +/- neuroimaging, EEG</i>); Cardiac: MI/CHF (<i>Troponin, CK, ECG +/-BNP</i>); Resp: asthma/COPDe, PE (<i>+/-CTPE</i>); Abdo: <i>exam</i> ; Derm: <i>rashes</i> ; MSK: <i>pain</i>

Delirium: Management

ABCs, consider thiamine, naloxone, O2, glucose depending on acuity

1. Treat the underlying causes and contributors

2. Supportive care (Non-Pharmacological)

- Multi-component intervention (see Prevention)
- **Pharmacological intervention not effective as treatment**
 - Systematic reviews have found statistically significant benefit to melatonin, ramelteon, and dual-orexin receptor antagonists, but low quality evidence and not in guidelines

3. Safety (Non-Pharmacological & Pharmacological)

- Address safety/behavioural disturbances to prevent complication
- Antipsychotics only if i) danger to self/others, ii) distressing psychosis, or iii) preventing medically necessary care
 - When needed: Low dose, short duration. Haldol or atypical antipsychotic 1st line¹⁰

1. NICE 2023 Delirium guidelines. 2. Burton et al. Cochrane. 2021 3. Burry et al. Cochrane. 2018. 4. Agar et al. JAMA Int Med. 2017. 5. Nikooie et al. Ann. Med. 2019. 6. CCSMH 2014 Delirium Guidelines 7. J Psychiatr Res 2021; Jan 133:181-190. 8. Michaela de Oliveira Crit Care Med. 2025. 9. Reppas et al. CMAJ 2023.

Delirium: Pharmacologic Strategies

Medication	Dose	Route	Sedation	Risk of EPS	Anticholinergic activity	QTc prolongation
Haloperidol	0.25-0.5 mg q4h	PO, IM, IV	Low	High	Low	High (esp if IV)
Loxapine	10 mg TID	PO/IM/IV	High	Moderate	Moderate	Moderate
Risperidone	0.25-0.5 mg daily to BID	PO, PO melt	Low	High	Moderate	Moderate
Olanzapine	2.5-5 mg daily to BID	PO, PO melt, IM, IV	High	Moderate	High	Moderate
Quetiapine*	6.25-25 mg daily to q8h	PO	High	Low	Moderate (if >100 mg/day)	Moderate
Aripiprazole	5-10 mg BID	PO	Low	Low	Low	Low
Lorazepam**	0.5-1 mg q4h	PO, SL, IM, IV	High	Low	Low	Low
Trazodone	25-150 mg qHS	PO	High	Low	Low	Moderate

*Preferred agent if Parkinson's or Dementia with Lewy Bodies.

** Preferred agent if EtOH withdrawal, NMS/serotonin syndrome. 2nd line if Parkinson's or DLB

BONUS

Read on own

A note on Peri-Op & ICU delirium

Preventing peri-operative delirium (POD):

- Risperidone after cardiac surgery (but not other surgeries) is associated with reduced risk of POD¹
- Dexmedetomidine intra-operatively for all surgeries may reduce risk^{2,3}
- Use of complimentary anesthesia (i.e. acetaminophen, fascia iliaca block, NSAID) reduces POD³
- Insufficient evidence for bispectral index monitoring intra-operatively⁴

ICU-related delirium:

- Eye masks and ear plugs are effective for prevention⁵
- Use of dexmedetomidine (versus other sedatives) for sedation of mechanically vented patients associated with reduced delirium risk (and ↑ bradycardia risk).^{6,7}
- Routine use of melatonin may ↓ prevalence of delirium in ICU⁸

1. Igwe et al J Clin Anesth, 2020. 2. Xin et al. Int J Geri Psych, 2021. 3. Subramaniam et al. JAMA, 2019. 4. Wildes et al. JAMA, 2019. 5. Litton et al. Critical Care, 2016. 6. Barbateskovic et al. BMJ Open, 2019. 7. Lewis et al. Intensive Care Med, 2022. 8. Lewis et al. SCCM Guideline Update. Crit Care Med, 2024.

III: Depression



Key Guidelines:

1. Canadian Coalition for Seniors Mental Health (CCSMH) Canadian Guidelines on Prevention, Assessment and Treatment of Depression Among Older Adults. 2021.
2. Canadian Network for Mood and Anxiety Treatments (CANMAT) 2016 Clinical Guidelines for the Management of Adults with Major Depressive Disorder: Section 6. Special Populations: Youth, Women, and the Elderly.
3. Not covered: Canadian Coalition for Seniors Mental Health (CCSMH) Canadian Guidelines on Assessment and Treatment of Anxiety Among Older Adults. 2024.

Depression: diagnosis & screening

- Diagnostic criteria same for older and younger adults: 5 or more in total of MSIGECAPS, need **anhedonia** or **low mood**
- **Differences in older adults:**
 - Chief complaint may be somatic or cognitive (“Pseudodementia” is the cognitive symptoms of depression d/t decreased frontal activity)
 - More complexity (eg, other chronic conditions like dementia or comorbidities causing poor concentration, low energy, low appetite, or even depression itself)
 - Attempt suicide less, but higher completion rate
- **Screening tools:**
 - Patient Health Questionnaire 9 (PHQ-9) Sensitivity 74-100%, specificity 53-98%¹ for score >10/27
 - PHQ-2 (first two questions of the PHQ-9) Sensitivity 86%, specificity 78%¹ 86% for score >2/6
 1. During the past two weeks, have you often been bothered by feeling down, depressed or hopeless?
 2. During the past two weeks, have you often been bothered by little interest or pleasure in doing things?
 - Geriatric Depression Scale (GDS), 15 item Sensitivity 74-100%, specificity 53-98%

Depression: management

Non-pharmacologic treatment:

- Group-based intervention
- Physical activity
- Cognitive behavioral therapy, problem-solving therapy

Advanced therapies:

- Electroconvulsive therapy (ECT) 2nd line after fail ≥ 1 adequate antidepressant trial, or 1st line if “high risk” (SI, psychotic features)
- rTMS (transcranial magnetic stimulation) can be considered for patients with unipolar depression who have failed 1 antidepressant (but do not use if already failed ECT)

Depression: management

Pharmacologic treatment¹:

- 1st line: sertraline, duloxetine
- 2nd line: citalopram, escitalopram (QT prolongation risk with both)
- Other appropriate therapies: venlafaxine, mirtazapine, bupropion, vortioxetine
- 3rd or 4th line: **Tricyclic antidepressants** (generally try to avoid! BEERS list med)
- Avoid: **paroxetine** (anticholinergic), **fluoxetine**, **MAO-i**
- Longer trial of therapy for effect (10-12 weeks)
- Reassess q1-2 weeks for drug side effects (hyponatremia with SSRIs, SNRIs), dose titration
- Augment (aripiprazole > bupropion) or switch if partial response after adequate trial (Minimum 4 weeks at maximally tolerated dose)

IV: Falls



Key Guidelines:

1. Montero-Odasso et al. 2022. World guidelines for falls prevention and management for older adults: a global initiative.
2. Sarmiento & Lee 2017 CDC STEADI Initiative. **2024** update for primary care.
3. AGS/BGS/AAOS Clinical Practice Guidelines. Falls Prevention. 2011 update.
4. Ganz et al 2007. JAMA Rational Clinical Exam: Will my patient fall?

Canadian Task Force for Preventive Health Care to release a Falls Plan in **2026**

Falls

Fall: An unexpected event in which an individual comes to rest on the ground, floor, or lower level

Recurrent falls: 2 or more falls in prior 12 months

***NB – “mechanical falls” – don’t use this nomenclature!**

Non-specific, obscures the multifactorial nature of falls → missed opportunities

- Falls are COMMON: 33% of people >65 years have a fall per year, ~10% have a major injury such as fracture¹
- Multiple systems (sensory [vision, vestibular, proprioception], motor, cognition, motor, perfusion) must be intact to stay upright

1. Screen Annually

3 Key Questions (3KQ)

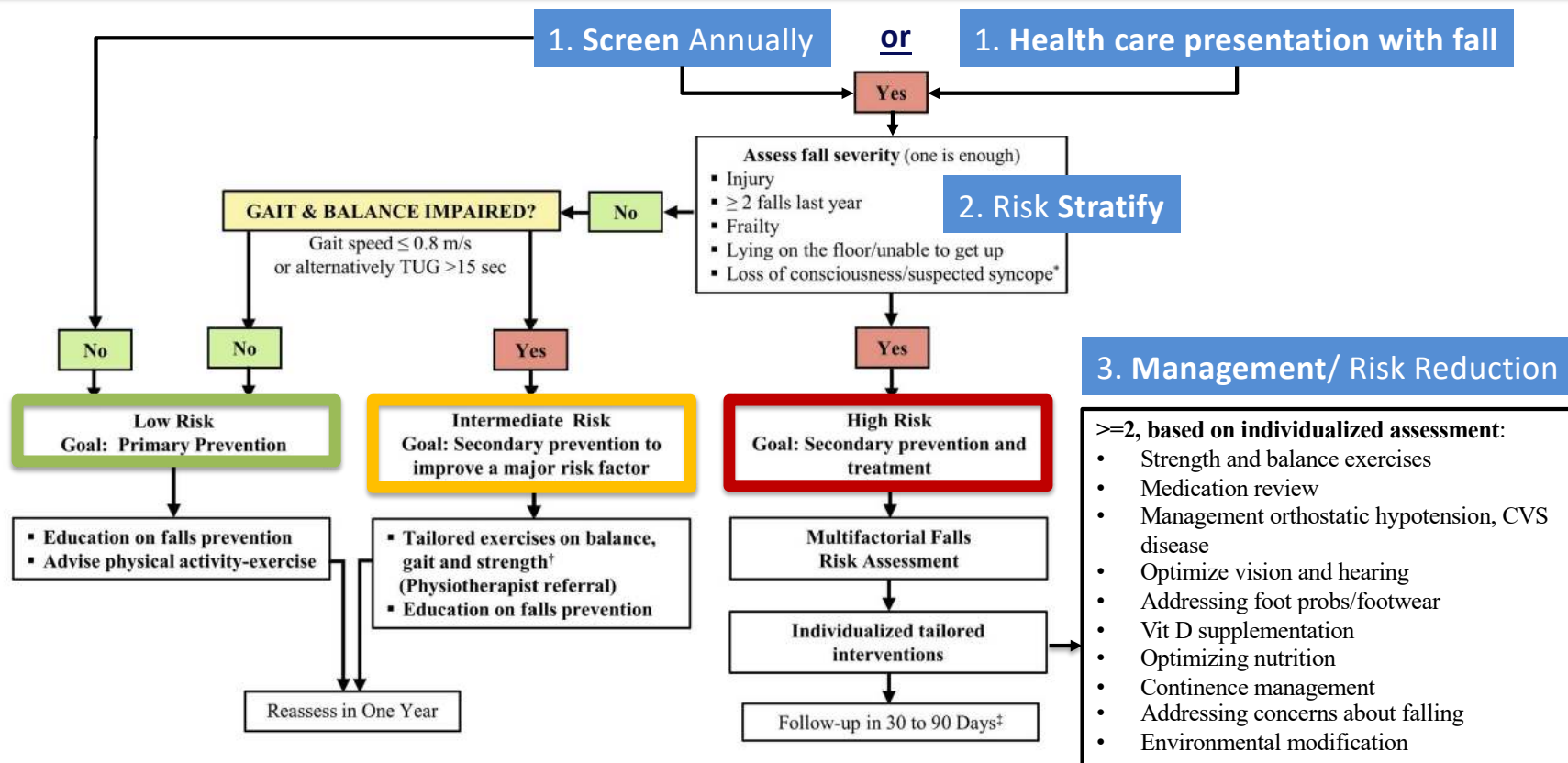
- 1) Have you fallen in the past year?
Additional questions to increase SEN:
- 2) Do you feel unsteady when standing/walking?
- 3) Do you worry about falling?

2. Risk Stratify

3. Management / Risk Reduction

Montero-Odasso et al. 2022
N Engl J Med. 2020 Feb 20;382(8):734-743.
J Am Geriatr Soc 2025 Jul;73(7):2019-2028.

Falls Overview : New Guidelines



Approach to Falls

1. What happened?

- What happened before? (Gait changes? What were they doing? Where? Prodromes?)
- What happened during? (Length of lie, How did they get up, Seizure?)
- What happened after? (Trauma, Fracture, Anxiety)

2. Why? Risk factors (see modifiable RF slides)

- Intrinsic: Head to toe, don't forget Geriatric ROS (incontinence, vision, hearing, nutrition, pain, cognition, mood)
- Extrinsic: Environment (rugs, railings, grab bars, slippery surfaces), footwear, restraints

3. Why now? (acute illness, delirium, seizure, syncope, neurodegenerative - PD, PSP, MSA, NPH)

4. How do we prevent the next fall?

- Assessment and management
- Patient perceptions of falls, risk, prevention, goals. Screen for concern for falling.

5. How do we prevent injury?

- Bone health
- Indication for blood thinner

Would you stop anticoagulation for stroke prevention? NO.

Patient would need to fall 295 times/year for risk of SDH to outweigh benefit for stroke prevention (warfarin data). Mitchell 2023. Eur Geriatr Med

BONUS
Read on own

Physical Examination/ Investigations



1. Vitals: Orthostatic
2. CVS: AS, arrhythmia, carotid bruit, **ECG**
3. [Routine Resp/Abdo – in acute setting to rule out illness as precipitant]
4. Neuro exam: cranial nerves - Vision screen, motor (tone, strength, reflexes), sensory (primary and secondary), cerebellar (finger-nose, heel shin), Parkinsonism
5. MSK exam: foot exam, knee and hip exam, osteoporosis
6. Gait and balance assessment: At minimum, **Gait Speed** or Timed Up and Go and balance screen (tandem gait, 1 leg stand)
 - Assess appropriate use of gait aid if applicable
6. Cognitive Testing: MMSE (**cognitive impairment** [LR 13])
7. Vision and hearing screen (at least in the past 1-2 years)

New decision tools with >95% sens for clinically important intracranial bleeding: CMAJ 2023; 195(47):E1614-E1621

Additional investigations should be tailored to findings on Hx & P/E

Montero-Odasso et al. 2022
Jepsen et al 2022 BMC Geriatr.

Falls Management

- All Risk Groups
 - **Education** on falls & fracture prevention including physical activity, lifestyle habits and adequate nutrition & vitamin D intake: All risk groups
- All Risk Groups (but especially Intermediate/High Risk)
 - **Exercise:** Tailored intervention targeting balance gait and strength training (multicomponent), by a trained professional, progressive intensity and then maintained
 - Including Tai Chi and/or individualized resistance training if possible
 - For individuals who are unable to get up on own, functional training is suggested
 - 3 times / week for at least 12 weeks
- High Risk Group
 - **Multifactorial Multidisciplinary Assessment & Individualized Fall Prevention Plan:** Take into account patient's priorities, beliefs and resources; Target modifiable RFs (Read On Own)
- **Bone Health:** Osteoporosis work-up/management (covered in Endocrinology lecture).

Vitamin D is recommended for *fracture* prevention, but no high-quality evidence in *falls* prevention. Supplement if **vit D-deficient** or **osteoporosis**.

Montero-Odasso et al. 2022
Endocrine Society Guideline 2024

BONUS

Read on own



Falls: Modifiable Risk Factors

	Recommendations
Medications	Med review and deprescribing using standardized tool (STOPP/START, STOPPFall, FORTA, Beers Criteria). RR 0.34 for withdrawal of psychotropic; benzo LR +27 for falls in one study
Cardiovascular/ Ortho Hypotension	Assess for and treat underlying cardiac conditions and orthostatic hypotension. Unexplained falls syncope equivalent when considering additional cardiac Ix
Balance and Strength	Individualized exercise program with balance and functional exercises 3 times per week for at least 12 weeks. Include Tai Chi and/or additional individualised progressive resistance strength training. (RR 0.68-0.85 with focus on balance & strength training.). Exercise benefit is less durable in LTC residents.
Environmental hazards	Assessment and modification of home environment considering capacities and behaviours by trained clinician (RR 0.69 for OT home safety assessment; footwear RR 0.64, non-slip boots for ice RR 0.42)
Nutritional Status	Assess nutritional status including vitamin D and supplement as appropriate (Expert opinion - conflicting evidence of vitamin D supplementation on falls but indicated for bone health)
Vision	Inquire and assess vision, refer to specialist as appropriate (RR 0.66 for 1 st cataract repair)

BONUS

Read on own



Falls: Modifiable Risk Factors

	Recommendation
Hearing	Inquire and assess hearing, refer to specialist as appropriate (expert opinion)
Vestibular	Inquire about, investigate, & manage dizziness, refer as appropriate (expert opinion)
Concern for falling	Screen with validated tool: Falls Efficacy Scale International (FES-I) or Short FES-I. Manage: CBT/ exercise / OT to reduce fear of falling (meta-analyses evidence of benefit)
Mood	Inquire about, assess, manage/refer as appropriate. (Expert opinion)
Pain	Assess and treat pain (expert opinion)
Urinary Sympt	Inquire about and manage/refer for urinary symptoms as appropriate (expert opinion)

Highlights of JAMA RCE “Will My Patient Fall” *Ganz 2007*

- Age > 65, 27% pretest probability
- Fall in past year (LR 2-3)
- Gait/balance abnormal (LR 2)
- Visual impairment (LR 1.5-2)
- Unable to rise from chair of knee height without use of arms (LR 4.3)
- Decreased ADL (LR 1.9)
- Unable to do tandem walk 2m (LR 2.4)
- Dementia (LR 17, 1 study)
- Evidence of prior stroke on exam (LR 17)
- Medications/ polypharmacy **Benzo use**
LR 27 in 1 study

Kruisbrink et al *Gerontologist* 2021, Chua et al. *J Adv Nurs* 2019, Coninck et al *JAGS* 2017, Montero-Odasso et al. 2022 Kwan and Straus. *CMAJ*. 2014., Tricco et al. *JAMA*. 2017. Ganz et al *JAMA* 2007, Guirguis-Blake et al. *JAMA*. 2018.

Falls Management in LTC

- All LTC residents should be considered high risk for falls
- Similar management: multifactorial assessment, address modifiable risk factors, nutritional optimization (Ca, protein, Vit D), Exercise training where feasible and safe

Do NOT use physical restraints (Geri chairs, lap-belts) as a measure for falls prevention (actually shown to increase falls) Castle et al 2009, Capezuti JAGS 1996, Capezuti JAGS 2004

Hip Fracture Prevention for Older Adults in LTC

Intervention	Hip fracture reduction
Multifactorial intervention: exercise, medication review, assess environmental hazards, assistive devices, manage incontinence	10 per 1000 treated
Hip protectors	4 per 1000 treated
Dietary calcium 1200 mg/d, vitamin D (800-2000IU) supplementation	5-15 per 1000 treated
Consider bisphosphonate/denosumab (if prognosis >1 year)	25 per 1000 treated

V. Disease Management in Older Persons



Normal Aging – An internist's summary



Function should not be impaired with normal aging alone

System	Change
Cognition	Preserved long-term functions (learned skills, vocabulary) Decreased short-term functions (attention, processing speed, short-term recall/working memory). Dementia is not normal!
CVS	Hypertension, widening pulse pressure, risk of HFpEF (thick stiff LV), decreased number SA node /pacemaker cells (risk of bradyarrhythmia, ↓ maximal HR and therefore cardiac output with exercise decreases)
Respiratory	Loss of elastic recoil, ↑ closing volume, ↓ chest wall compliance). This causes ↑ RV and dead space, ↓ VC, FEV1, DLCO → less reserve, greater V/Q mismatch.
Renal	Decreasing eGFR (1ml/min/yr after age 40), Cr production also declines, decreased functional reserve (higher risk of AKI), decreased 1-alpha hydroxylase activity = reduced active vitamin D
GU	Increased volume at first desire to void, decreased bladder capacity, detrusor activity, and sphincter tone Men: increase prostate size, Women: urethral shortening and mucosal atrophy
GI	Constipation, decreased phase 1 liver metabolism (oxidation, reduction, hydrolysis); Phase II (conjugation) usually preserved.
MSK	↓ muscle and bone mass, ↓ muscle force, ↓ processing speed;
Neuro	Slowed reflexes, reduced sensation; ↓ Balance: ↓ processing speed and speed/amplitude of postural reflexes, ↓ muscle and joint proprioception, ↑ truncal sway, ↓ efficiency of the vestibular system



Hypertension in Older Adults

- Age ≥ 75 is a criterion for intensive management
 - Target sBP < 130 per Hypertension Canada 2025¹, “exceptions: goals of care, frailty, fall risk, and orthostatic hypotension”
 - sBP target <120 in SPRINT acknowledged in guideline
 - Initiate pharmacotherapy when SBP 130-139¹

Exclusion criteria for SPRINT trial:

- T2DM, previous stroke, eGFR < 20, protein >1g/day
- Orthostatic hypotension with SBP < 110mmHg
- Dementia, prognosis <3 year, institutionalized

Recent relevant literature leans towards tighter control

- ESPRIT Lancet 2024²: SBP <120 vs 140 in high-risk older adults, including with stroke and diabetes, found small benefit NNT 71, but NNH renal disease 78
- Cochrane review 2024 (n = 16732; mean age 70.3)³: sBP <140 vs <150-160 reduces stroke and total serious CV events, and likely does not increase withdrawals due to adverse effects

But when you also consider frailty...

- Meta-analysis⁴ finds no benefit with tighter control of sBP <140 vs >140 in frail persons

What about LTC residents? Common practice to “liberalize BP targets” but...

- RETREAT-FRAIL 2025⁵: decreasing antihypertensives in frail LTC patients with baseline sBP <130 did not change all-cause mortality/major CV events/falls/fractures/functional decline

Diabetes in Older Adults*



Diabetes Canada guidelines:

- Avoid sulfonylureas (eg, glyburide) due to hypoglycemia risk
- Generally: metformin, then DPP-4 inhibitors or GLP-1 (known side effects of weight loss), then SGLT2 (may be at higher risk of dehydration) then long-acting insulin
- A1c targets should scale based on functional status
 - If below target, de-intensify therapy

BEERS criteria 2023:
Recommends against Sliding Scale insulin for outpatients

Functionally independent	A1c $\leq$ 7.0%
Functionally dependent	A1c 7.1 - 8.0% (<8.0% if lower hypoglycemia risk)
Frail and/or dementia	A1c 7.1 - 8.5% (<8.5% if lower hypoglycemia risk)
End of life	avoid symptomatic hyperglycemia and hypoglycemia, CBG checks not indicated

* Older adult defined as >70 but with greater focus on functional status, as many older adults age 70+ are independent with > 10 year life expectancy and would benefit from usual management

During hospitalization



Avoid intensification of antihypertensives and diabetes regimens among patients over age 65 during hospitalization

- Intensification of antihypertensives not associated with reduced cardiac events or improved BP control, but increased 30 day readmission and adverse events¹
- Intensification of diabetes medications not associated with reduced A1c or reduced hyperglycemia, but increased risk of hypoglycemia within 30 days²

Primary prevention of CV events



- **Aspirin**

- Not for primary prevention of cardiovascular disease (ASPREE trial; increased mortality due to cancer deaths in intervention arm)¹
- Still indicated for secondary prevention
- Not indicated for vascular dementia without clinical stroke history

- **Statin >75 years of age**

- CCS 2021 Lipids “accumulating evidence suggests continued benefits of lipid-lowering for primary prevention in older adults”²
- American Geriatrics Society 2025 “for LDL 1.8-4.9, may be reasonable in those without life-limiting illness” “shared decision-making”³
- AHA and USPSTF suggest individualized choice >75 yrs⁴ and >76 yrs⁵, respectively

Statin therapy tips

- Think twice about starting if life expectancy <2.5 yrs (NNT 100)⁶
- Don't D/C without cause: NNH (MACE) from statin discontinuation in older adults: 112 (primary prevention), 77 (secondary prevention)⁷

1. NEJM. 2018 Oct;379(16):1519-1528. 2. Can J Cardiol. 2021 Aug;37(8):1129-1150. 3. J Am Geriatr Soc. 2025 Jun;73(6):1674-1696. 4. Circulation. 2019 Jun;139(25):e1046-e1081. 5. JAMA. 2022 Aug;328(8):746-753. 6. JAMA Intern Med. 2021 Feb; 181(2):179-185. 7. JAMA Netw Open. 2021 Dec;4(12):e2136802

Sleep in Older Adults

Intervention	Notes
<i>Psychological</i>	
Cognitive behavioral therapy for insomnia (CBT-I)	First line ^{1,2} , favorable risk/benefit profile. Brief behavioral treatment for insomnia (BBT-I) is an abbreviated form of CBT-I, also shown to be efficacious.
<i>Pharmacological "Pick Your Poison"</i>	
Melatonin	Insufficient evidence but well tolerated. Use in REM sleep behavior disorder (RBD). *Ramelteon is a melatonin-receptor agonist, not available in Canada.
Dual Orexin Receptor Antagonist (DORAs): suvorexant, lemborexant, daridorexant	All showed efficacy in improving sleep parameters, but will have daytime somnolence, reports of nightmares, sleep paralysis, depression. ³ Falls reported but considered "not related to treatment", practically still proceed with caution.
Mirtazapine	2025 MIRAGE Trial ⁴ showed reduction in insomnia severity in older adults, with mild but clinically relevant adverse events (daytime somnolence, confusion). Beer's List insufficient evidence to recommend. Paradoxical decrease in sedative effect with dose > 15mg!
Trazodone	BEERS: insufficient evidence to recommend.
Doxepin	<6mg has low quality evidence for improving sleep parameters. Not in Beer's List.
Benzodiazepines, Z-drugs, Sedating anti-histamines	Don't use. impair psychomotor performance and memory the next day, falls, rebound insomnia. Some evidence for short-course but ultimately risk > benefit in older adults.

Polypharmacy



Key Resources:

- [Medication Appropriateness Index](#)
- [STOPP/START](#)
- deprescribing.org
- [Flockhart Table](#)
- Scott et al. JAMA Intern Med. 2015.
- American Geriatrics Society Beers Criteria. Updated 2023.
- Alternative Treatments to Selected Medications in the Beers Criteria. 2025.

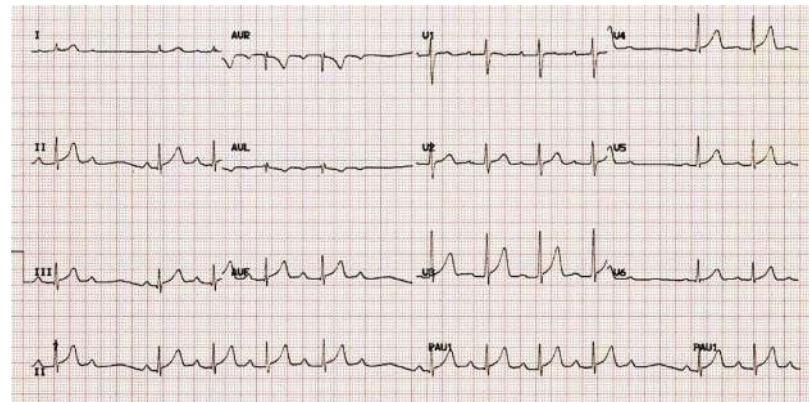
MCQ #4 2026

A 78-year-old man is brought to clinic for evaluation of dizziness and poor appetite. Over the past month, he has had several near-syncope episodes and 5 pounds of unintentional weight loss. He also reports loose stools twice per week. His medical history includes hypertension, dyslipidemia, type II diabetes mellitus, and Alzheimer dementia. Supine blood pressure is 122/66 mmHg, HR 54/min; standing blood pressure is 116/62 mmHg, HR 58/min. HbA1c 6.5%. ECG as shown.

Medications: Amlodipine 10 mg daily, Atorvastatin 20 mg nightly, Metformin 1000mg po BID, Sertraline 50 mg daily, Ramipril 10 mg daily, Rivastigmine 6 mg twice daily

Which of the following medications is the most appropriate medication to deprescribe next?

- A. Rivastigmine
- B. Amlodipine
- C. Sertraline
- D. Metformin



<https://litfl.com/>

Polypharmacy

Definition: More medication than is clinically indicated or warranted.

Various cut-offs depending on source: >5 medications, >9 medications, etc.

- Why does it matter:
 - more meds + frail older adult = more adverse events
 - 50% of elderly (>65) w/ 5+ meds experience Adverse Drug Event
- Avoid prescribing cascade:
 - Dementia → rx donepezil → urinary incontinence → rx oxybutynin → worse cognition
- Resources to help identify medications for deprescribing:
 - Beers' Criteria (list by pharmacologic effect), STOPP/START criteria, Deprescribing.org, Medication Appropriateness index, Flockhart Table

Approach to Deprescribing

- 1. Determine all drugs (prescribed, OTC, supplements, substances) currently taking (or not taking) and reasons for each**
 - Requires BPMH (multiple information sources) consider pharmacy consult
- 2. Consider overall risk of drug-induced harm**
 - Number of drugs, high risk drugs (ex. benzo), prior/current toxicity,
 - Patient characteristics: age >80, renal function, cog impairment, non-adherence, substance use
- 3. Assess each drug's eligibility to be discontinued**
 - No valid indication, no longer indicated, non-adherence with no negative consequence, part of prescribing cascade, inappropriate dose, lack of clinical benefit, harms (actual or potential) outweigh benefit (current or future), preventative and unlikely to benefit given remaining lifespan, patient preference
- 4. Prioritize drugs for discontinuation**
 - Greatest harm, least benefit
 - Least risk of withdrawal or disease rebound
 - Patient buy-in
- 5. Implement a discontinuation regimen and monitor patients**
 - 1 drug at a time, wean to prevent withdrawal, document reasons and inform other healthcare providers

Potentially Inappropriate Medications

American Geriatrics Society offers the Beers' Criteria, last updated **2023**:

Category	Examples
PIMs to avoid	Warfarin, Rivaroxaban , Sliding scale insulin, anticholinergics, sulfonylureas, benzodiazepines, sedatives, etc.
PIMs to be used with caution	Dabigatran for AF, TMP-SMX with ACEI
PIMs due to drug-disease interaction	Syncope and ChEIs/TCAs/alpha-blockers, Parkinson's disease and metoclopramide, falls and antiepileptics
PIMs due to drug-drug interactions	Opioids and sedatives, opioids and gabapentinoids, warfarin + SSRI, Li +ARB/ACEi
PIMs to avoid based on renal clearance	Baclofen , Spironolactone, ciprofloxacin, enoxaparin, gabapentin, etc.
Anticholinergics	Amitriptyline, dimenhydrinate, hydroxyzine, tolterodine, paroxetine

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Which of the following medications is the most appropriate medication to deprescribe next?

- A. Rivastigmine**
- B. Amlodipine
- C. Sertraline
- D. Metformin

The correct answer is A. ChEIs' main side effects are related to parasympathetic activation. Here, recognize the constellation of side effects: bradyarrhythmias, syncope, and gastrointestinal side effects including anorexia, nausea, and diarrhea. While amlodipine can contribute to hypotension and cause dizziness, his blood pressure is WNL, sertraline does not typically present with heart block, metformin would be reasonable to decrease as HbA1c is too tightly controlled but not the next best agent.

VI. Vaccinations in Older Adults

- Tdap: one booster per lifetime
- Td (tetanus, diphtheria): every 10 years
- Influenza: annually for >65yo
- **RSV (2025 Update)**
 - RSVPreF3, RSVpreF, mRNA-1345 once if >60yo or in institutional care (LTC/Retirement)
 - RSVPreF3 for 50-59yo if high risk for RSV
- COVID 19 mRNA vaccination – fall booster recommended (see Bonus slide)
- Pneumococcal: Prevnar + Pneumovax (8 wks later) at 65yo
- Zoster: recombinant subunit adjuvant vaccine (Shingrix), as of October 2020; 2 doses 2-6mo apart



Choosing Wisely Geriatrics Recommendations



1. Don't use **antimicrobials to treat bacteruria** in older adults unless specific urinary tract symptoms are present.
2. Don't use **benzodiazepines or other sedative-hypnotics** in older adults as first choice for insomnia, agitation or delirium.
3. Don't recommend **percutaneous feeding tubes** in patients with advanced dementia; instead offer oral feeding.
4. Don't use **antipsychotics** as first choice to treat behavioural and psychological symptoms of dementia.
5. Avoid using medications known to cause hypoglycemia to achieve **hemoglobin A1c <7.5%** in many adults age 65 and older; moderate control is generally better.
6. Avoid **physical restraints** to manage behavioural symptoms of delirium and dementia
7. Don't continue **ChEi or memantine** without periodic reassessment for perceived benefits (cognitive, functional, behavioural) and adverse effects, and consider deprescribing when risks outweigh benefits
8. Don't prescribe a medication without conducting a **medication reconciliation review** and consider opportunities for deprescribing at interfaces of care
9. Don't do **in-person visits** when a **virtual visit** is clinically appropriate for older adults, when preferred by the patient or family, or where travelling to an in person appointment proves difficult.
10. Don't **discard unused and expired medications** (prescribed or non-prescribed) and natural health products down sinks, toilets, or the trash. Instead, bring them to a designated collection site for proper disposal.
11. Don't **recommend/order investigations or interventions** before considering a patients' expected trajectory of health and life expectancy, and discussing their preferences, values and goals of care.
12. Don't discard **adaptive equipment, mobility devices, and wheelchairs** that are no longer being used, but consider recycling or reusing them.

New Apr 2025

Questions?

BONUS SLIDES

Good luck on your exam!



- COVID-19 and Geriatrics
- Functional Assessment/Frailty
- Normal physiology of aging
- Medications and polypharmacy
- Nutrition and weight loss
- Incontinence and Constipation
- Extra MCQs

COVID-19 & Geriatrics

- COVID-19 impacts older adults disproportionately—from severe illness and hospitalization to increased risk for death.
 - People aged 80 or over have a more than 20 fold increased risk compared with 50-59 year olds
- National Advisory Committee on Immunization 2025
 - COVID-19 mRNA vaccination recommended for all older adults
 - Second dose recommended for those who are previously vaccinated and are at increased risk of severe COVID-19 disease
 - Adults 80 years of age or older;
 - Adult residents of long-term care homes and other congregate living settings for seniors

Functional assessment

Basic (BADLs)	Instrumental (IADLs)
Dressing	Medication management*
Grooming	Finances *
Toileting	Driving *
Ambulation/transfers	Technology (computer/smartphone) use
Eating	Cooking
Bathing	Cleaning
	Shopping
	Laundry

* Highest level / often first to be impaired

Rockwood Clinical Frailty Scale

CLINICAL FRAILTY SCALE

	1	VERY FIT	People who are robust, active, energetic and motivated. They tend to exercise regularly and are among the fittest for their age.
	2	FIT	People who have no active disease symptoms but are less fit than category 1. Often, they exercise or are very active occasionally , e.g., seasonally.
	3	MANAGING WELL	People whose medical problems are well controlled, even if occasionally symptomatic, but often are not regularly active beyond routine walking.
	4	LIVING WITH VERY MILD FRAILTY	Previously "vulnerable," this category marks early transition from complete independence. While not dependent on others for daily help, often symptoms limit activities . A common complaint is being "slowed up" and/or being tired during the day.
	5	LIVING WITH MILD FRAILTY	People who often have more evident slowing , and need help with high order instrumental activities of daily living (finances, transportation, heavy housework). Typically, mild frailty progressively impairs shopping and walking outside alone, meal preparation, medications and begins to restrict light housework.

	6	LIVING WITH MODERATE FRAILTY	People who need help with all outside activities and with keeping house . Inside, they often have problems with stairs and need help with bathing and might need minimal assistance (cuing, standby) with dressing.
	7	LIVING WITH SEVERE FRAILTY	Completely dependent for personal care , from whatever cause (physical or cognitive). Even so, they seem stable and not at high risk of dying (within ~6 months).
	8	LIVING WITH VERY SEVERE FRAILTY	Completely dependent for personal care and approaching end of life. Typically, they could not recover even from a minor illness.
	9	TERMINALLY ILL	Approaching the end of life. This category applies to people with a life expectancy <6 months , who are not otherwise living with severe frailty . (Many terminally ill people can still exercise until very close to death.)

SCORING FRAILTY IN PEOPLE WITH DEMENTIA

The degree of frailty generally corresponds to the degree of dementia. Common **symptoms in mild dementia** include forgetting the details of a recent event, though still remembering the event itself, repeating the same question/story and social withdrawal.

In **moderate dementia**, recent memory is very impaired, even though they seemingly can remember their past life events well. They can do personal care with prompting.

In **severe dementia**, they cannot do personal care without help.

In **very severe dementia** they are often bedfast. Many are virtually mute.



DALHOUSIE UNIVERSITY

Clinical Frailty Scale ©2005–2020 Rockwood, Version 2.0 (EN). All rights reserved. For permission: www.geriatricmedicine.ca
Rockwood K et al. A global clinical measure of fitness and frailty in elderly people. CMAJ 2005;173:489–495.

Physical Examination Maneuvers



Examination	How to do it
Orthostatic hypotension	Lie down for 5 minutes, measure BP and HR supine, stand the patient, repeat BP and HR at 1 and 3 minutes. Positive if: SBP drops by 20mmHg, DBP drops by 10mmHg or patient is symptomatic.
Vision	Snellen chart (wall, or handheld)
Hearing	During the examination; can offer a pocket talker
Gait assessment	General description: symmetric or asymmetric, wide or narrow based, high or low step height Types of gait: antalgic, hemiplegic, ataxic, parkinsonian (shuffling, festination), steppage, spastic Difficulty standing without pushing off chair [LR 4.3 falls in men]
Timed Up and Go	Stand up, walk 3 metres, turn around, sit down. Normal <12s, World Falls Guidelines >15 is high falls risk.
Neuro/MSK	Motor (check tone, strength, reflexes), sensation (light touch, vibration, proprioception), cranial nerves, cerebellar examination (coordination), Parkinsonism

A note on elder abuse



- Comes in many types:
 - Physical
 - Verbal
 - Emotional
 - Sexual
 - Financial
 - Neglect
- Most common type of fracture seen in physical elder abuse is a facial fracture in maxillofacial region (mandible, zygoma, orbit, nasal bones)
- Mandatory reporting per provincial requirements
 - Required if abuse occurs in long-term care or retirement home (Ontario)

1. Rosen et al. Ann Emerg Med. 2020
2. Gironde et al. J Am Geriatr Soc. 2016 Aug;64(8):1651-5.
3. Khurana et al. J Am Geriatr Soc. 2024 Apr;72(4):1011-1022.

Normal Aging: Neuro Exam

- Cranial nerves:
 - Vision: ↓ accommodation, reduced contrast sensitivity, dark adaptation and depth perception
 - Hearing: Presbycusis – bilateral symmetric sensorineural hearing loss ↓ high frequency, cerumen impaction
 - Diminished olfaction
- Peripheral nervous system:
 - ↓ or absent ankle reflexes
 - ↓ vibration sense to ankle, proprioception
- Autonomic nervous system:
 - ↓ baroreceptor response, increase vascular tone/decreased elasticity (SBP>>DBP, wider pulse pressure)
 - ↓ parasympathetic innervation of detrusor muscle [increase post void residual, nocturia, incontinence]
- Gait:
 - 10-20% reduction in gait velocity and stride length. Increased stance width, increased time in double support phase.

Pharmacodynamics & Aging

“What drug does to body”

Minimal change w/ age, some ↑ sensitivity

- Additive: $A+B = AB$
 - ex: ASA & SSRI (for platelet inhibition)
- Antagonistic: $A+B < AB$
 - ex: oxybutynin & ChEI (for cholinergic effect)

Older adults more sensitive	Older adults less sensitive
Antipsychotics	Beta-blockers
Anticholinergics	Stimulants
Opioids	
Sedative hypnotics	

Sensitivity: experiences greater physiologic effect with the same dose

Pharmacokinetics & Aging

“What body does to drug”

Significant changes w/ age

- **Absorption:** relatively unaffected
- **Distribution:** Less muscle & water, more fat
 - Water soluble drugs – increased serum concentrations (i.e., lithium)
 - Fat soluble drugs – increased half-life (i.e., haloperidol)
- **Metabolism:** Reduced hepatic metabolism
 - Enzymatic transformations:
 - Demethylation
 - **Phase 1 (CYP450) oxidation (decreased ~30%)**
 - Phase 2 glucuronidation (less affected)
- **Elimination:** Reduced GFR – less clearance

Considerations in Med Review

Reason to stop or D/C	Selected examples
Non-compliance w/o negative health effect	Anti-HTN Oral diabetic medication
Takes incorrectly w/o negative effect or benefit	Puffers
Not indicated or relative contraindication	Statin, ASA for primary prevention Docusate CCB in pt with systolic heart failure
No longer indicated	Bisphosphonate after 5 yrs* Benzos, NSAIDs, PPIs ChEI (consider if fully dependent)
Inappropriate dose for geriatric patient	Digoxin Amitryptiline
Not aligning with goals of care, life expectancy	Bisphosphonate (<1 yr life expected) Warfarin

Adapted from Frank et al. Deprescribing for older patients. CMAJ 2014

Approach to Weight loss in older adults

Inadequate caloric intake	Poor absorption	Increased metabolic needs
• Normal age-related: decrease in grelin, slower gastric emptying, decreased effectiveness of peristalsis (+T2DM-associated gastric motility changes) • Cognitive changes: low mood, loneliness, psychiatric illness • Oral issues: GERD/acid taste, poor oral hygiene, poorly fitting dentures, dry mouth from medications, angular chelation • Dysphagia/odynophagia • Nausea: central or peripheral (GERD, constipation, med side effect) • Decreased mobility: decreased access to food, stamina to cook • Inactivity: reduced hunger drive • Food insecurity	• Inflammatory bowel disease • Celiac • Pancreatic insufficiency (*floating stools*) • Small intestinal bacterial overgrowth (*bloating, flatulence, watery diarrhea)	• Increased physical activity (wandering) • Endocrine: TSH, cortisol • Malignancy: up to date screening • Infection: TB, HIV
APPROACH: Organize into factors that affect caloric intake, absorption and metabolic use. Remember to check for medication side effects!		

Weight loss in older adults

Common causes of unintentional weight loss in elderly patients:

- Malignant disease (16%–36%)
- Psychiatric disorder (especially depression) (9%–42%)
- Gastrointestinal disease (6%–19%)
- Endocrine disorder (especially hyperthyroidism) (4%–11%)
- Cardiovascular disease (2%–9%)
- Nutritional disorders or alcoholism (4%–8%)
- Respiratory disease (~6%)
- Neurologic disorder (2%–7%)
- Chronic infection (2%–5%)
- Renal disease (~4%)
- Connective tissue disease (2%–4%)
- Drug-induced weight loss (medication side effects) (~2%)
- Unknown (10%–36%)

Urinary Incontinence

Simplified Bladder Physiology

- Sympathetic Nervous System (STORES!)
 - alpha adrenergic → constricts internal sphincter
 - beta adrenergic → relaxes detrusor
- Parasympathetic Nervous System (PEES!)
 - cholinergic → contracts detrusor

Most common UI in older women:
Urge or Mixed (~70%)

Types of Urinary Incontinence

- Stress
 - Involuntary leakage on stress/exertion/cough/sneezing
- Urge
 - Sudden, strong need to void and involuntary leakage
 - History of urine loss associated with urinary urgency +LR 4.2
- Mixed
 - Combination of stress and urge
- Overflow
 - Overdistension of bladder (obstruction, neurologic)
- Functional (Can't get there in time)
 - Cognitive, functional, mobility difficulty

Incontinence Assessment

- Work-up:
 - PVR rule out retention (if retaining, cystoscopy)
 - Med review/ comorbidity review
 - Caffeine/EtOH/ free fluid intake
 - Vaginal exam
 - Abdo exam/Abdo XR/DRE
 - Depression screen
 - Urinalysis
- Identify and treat causes/contributors
 - Medications: Diuretics, CCB/NSAID/Pregabalin – peripheral edema redistribution and nocturia at night, anticholinergic (retention → overflow), Sedating medications (functional) and MORE
 - Comorbidities: PD, dementia, depression, uncontrolled DM, sleep apnea, severe constipation (retention -> overflow), and MORE

For women with genitourinary symptoms of menopause, or recurrent UTIs :

- Pelvic floor physio
- VAGINAL ESTROGENS

Pick these BEFORE you pick other pharmacotherapy for overactive bladder or recurrent UTIs.

Vaginal estrogen is FIRST LINE in women with recurrent UTIs

Do not pick ORAL /SYSTEMIC/Transdermal Estrogens for genitourinary syndrome of menopause or vasomotor symptoms of menopause in elderly [Beers list, 2023]

Incontinence Management

- Non-Pharmacological (1st line treatment for all types):
 - LIFESTYLE modifications: limit fluids, caffeine and EtOH; weight loss; treat constipation; bladder training (cog intact), pelvic floor muscle training (cog intact), prompted/timed voiding (cog impaired), functional intervention training, or pessary (stress)
- Stress:
 - Pessary for prolapse, no pharmacotherapy, surgical options
 - Effective, but trials do not represent older adult population. Health Canada warning re: mid urethral sling due to complications.
- Urge:
 - **Beta-agonist (Mirabegron)**: s.e. headache, tachycardia afib, HTN, nasopharyngitis effectiveness low (<1/3), ++ side effects (on BEERs List to avoid)
 - **Anticholinergic**: s.e. sedation, confusion, dizziness, urinary retention, constipation,
 - On Beer's List. Fesoterodine is most studied with more favorable cognitive profile, oxybutynin worst
 - Botox into detrusor

Constipation

Underlying causes for constipation:

- medications: anticholinergics, opiates, iron, calcium, calcium channel blockers, NSAIDs
- immobility
- neurologic: Parkinson's disease, diabetes, stroke, spinal cord injury
- volume depletion
- metabolic: hypothyroidism, hypercalcemia, hypokalemia
- mechanical obstruction
- dementia or depression
- low dietary fibre

CONSTIPATION RED FLAGS:

- Family history of colon cancer
- Hematochezia
- Anemia
- Weight loss ≥ 5 kg in previous 6 months
- Positive result of fecal occult blood test
- Persistent constipation unresponsive to treatment
- Acute onset of constipation

Constipation Approach	Constipation Interventions
1. Determine predominant symptoms	Frequency, straining, incomplete evacuation
2. Identify possible secondary causes of constipation	• If alarm S&S are present, follow guidelines for colon cancer screening
3. Consider fecal impaction	• Consider abdominal XR or DRE to diagnose (especially if bedbound, advanced dementia) • Manual disimpaction if fecal impaction present
4. Optimize behavioural factors	• Seated position, knees at/above the level of hips • If moderate to severe cognitive impairment, allow time to toilet after am meal, to take advantage of gastrocolic reflex
5. Trial of dietary modifications (2-4 wks)	• <u>Gradually</u> increase fibre intake to 20–30 g/d from dietary (fruits, vegetables, legumes) or supplemental sources (psyllium, methylcellulose, calcium polycarbophil) • Not advised in a person who is immobile or bedbound, to avoid impaction or obstruction
6. Trial of previously preferred laxative agent (2-4 wks)	• The patient may prefer one agent over another from past experience
7. Trial of laxative agents from RCTs involving older people (2-4 wks)	• Polyethylene glycol 17–34 g/d • Lactulose 15–30 mL daily to twice daily
8. Trial of another laxative agent or combination of agents from different classes (2-4 wks)	Magnesium hydroxide 15–30 mg daily to twice daily; Docusate calcium 240 mg twice daily; Bisacodyl 5–10 mg/d orally or rectally; Sennosides, up to 68.8 g/d in divided doses; Enema or suppository
9. Consider referral	Geriatrician or gastroenterologist

BONUS MCQ (2025)

An 80-year-old retired truck driver is referred for cognitive assessment. His partner describes a 2-3 year history of worsening forgetfulness, reports he frequently repeats stories and forgets things he has been told. This year his partner took over managing finances and appointments due to missed bill payments and doctor's appointments. He drives but only to and from the grocery store in his rural town. He hasn't had any near-misses or accidents and his partner says they feel safe in the car with him. He never gets lost or loses concentration when driving. MoCA 22/30 with points lost on delayed recall, orientation and sentence repetition. What is the next best step in management?

- A. Ask him to stop driving. Re-evaluate him with Trails A and Trails B in six months time.
- B. Ask him to sign an agreement to stop driving pending completion an on-road driving assessment.
- C. Tell him to stop driving. Report to the Ministry of Transport only if he continues to drive.
- D. Tell him he must stop driving and report him to the Ministry of Transport for license suspension.

The correct answer is D. C could be correct in AB, BC, NS and QB where reporting is discretionary, but according to the CMA Drivers Guide (10th ED, 2023), 'Dx of dementia alone is inadequate to withdraw driving privileges but driving is contraindicated in people with severe dementia (not this patient) and people who cannot perform and ADLs or 2+ IADLs secondary to cognitive changes (this patient), as driving is considered the highest order IADL. See <https://drivinganddementia.ca> for more information.'

BONUS MCQ (2025)

A 73-year-old previously well woman is seen in clinic. Her family reports that over the past several years, she's had declining ability to multitask and has been missing bill payments. She recently got lost on a walk in her own neighborhood. She's also been talking to young children in the room that only she sees; this happens during church and has become very disruptive to her ability to attend weekly sermons. Review of systems is significant for kicking and punching in her sleep and slowing of her gait. What medication is best for treating her hallucinations so that she may continue attending church without disruption?

- A. Quetiapine 25 mg PO qhs
- B. Memantine 10 mg PO qd
- C. Risperidone 1 mg PO qhs
- D. Rivastigmine 1.5 mg PO bid

The correct answer is D. Story highly suggestive of dementia with Lewy Bodies. Cholinesterase inhibitors are 1st line for treatment of cognitive symptoms and hallucinations in people with alpha-synucleinopathies such as dementia with Lewy Bodies or Parkinson's disease with dementia (Grade A evidence for rivastigmine and donepezil). Memantine can be trialled if cholinesterase inhibitors fail/are not tolerated (Grade C). Antipsychotics are generally contraindicated, as people with alpha-synucleinopathies are exquisitely sensitive to sedating/NMS side effects. Quetiapine is best tolerated antipsychotic, but wrong because hallucinations in alpha-synucleinopathies respond better to cholinesterase inhibitors than antipsychotics.

BONUS MCQ (2025)

A 92-year-old lady is admitted to hospital from home where she ambulates independently with a walker. She's had four days of dysuria, polyuria and chills, and an unwitnessed fall on the day of hospitalization. She cries out in pain when she tries to sit up in bed for your exam and physiotherapy says they're unable to mobilize her due to hip pain. There are no deformities to her hips or knees on physical exam. Anterolateral and posterior XR of her pelvis reveal no fractures, but 24 hours after admission she is still unable to weight bear. What should be done next?

- A. Repeat pelvic XR on day 2 of admission.
- B. Non-contrast CT pelvis on day 2 of admission.
- C. Acetaminophen 1 g PO TID and hydromorphone 0.5 mg PO q2hr PRN to facilitate participation in rehabilitation.
- D. MRI pelvis on day 3 of admission.

The correct answer is B. According to two systematic reviews (Radiology 2020 Sep; 296:521, AJR Am J Roentgenol 2020 Sep; 215:559), 39% of patients with suspected hip fracture but no surgical fracture on initial hip x-ray had surgical hip fracture identified on MRI. MRI is gold standard (Sens 100%, spec 99%) vs CT hip (Sens 79%, spec 91%) but we know that morbidity and mortality increase if surgery delayed >24 hrs (JAMA 2017 Nov 28;318(20):1994-2003) so we do CT first, pursue MRI only if CT negative.

BONUS MCQ (2025)

A 72-year-old woman is admitted for management of urosepsis. She has idiopathic Parkinson's disease treated with levodopa-carbidopa 2 tabs TID for the past two years. This has been continued throughout her hospitalization. You are called to evaluate her because she has injured two nurses via punching during peri-care and has repeatedly pulled out IVs providing necessary antibiotics and IV fluids. She is hallucinating insects flooding her room and this is very distressing for her. Her family says all of these symptoms are new. You complete a comprehensive delirium work-up and implement a multidisciplinary non-pharmacologic delirium management intervention with your allied health colleagues. What do you prescribe reduce risk of harm to staff and facilitate medically necessary care?

- A. Risperidone 0.5 mg PO/SL BID PRN.
- B. Quetiapine 12.5-25 mg PO TID PRN.
- C. Haloperidol 0.5-1 mg PO/IV TID PRN.
- D. Lorazepam 0.5-1 mg PO/IM/IV TID PRN.

The correct answer is B. Haloperidol is first line according to CCSMH 2014 delirium guidelines, but not in people with Parkinson's disease who are at very high risk of developing EPS secondary to haloperidol and risperidone administration. Per the 2019 Canadian Parkinson's Disease guideline (CMAJ), in people with Parkinson's and dementia with Lewy Bodies, choose quetiapine as 1st PRN option. Lorazepam is 2nd line.

BONUS MCQ (2025)

A 65-year-old woman with well-managed hypertension and hypothyroidism presents to ED with a 7-week history of progressive left-sided clumsiness, slow halting speech, confusion and disorientation. On exam, she was disoriented and amnesic with left-sided weakness in arm extension and leg flexion, hyperreflexia and increased tone. Which of the following findings is most suggestive of NMDAR-antibody disease as an underlying etiology?

- A. Periodic biphasic or triphasic sharp wave complexes on EEG
- B. Elevated 14-3-3 protein in CSF
- C. Startle myoclonus
- D. Ovarian teratoma on CT pelvis

The correct answer is D. All the other listed findings, including the patient's case presentation, are very in keeping with CJD. NMDAR-ab disease tends to occur in young people (often females) with short-term memory loss, psychiatric symptoms, insomnia, dysautonomia, seizures, hyponatremia; ovarian teratoma is found in 60% of affected female patients.

BONUS MCQ (2025)

An 82-year-old woman is brought to ED by her family for strange behaviour. She is typically independent in IADLs and ADLs with no marked cognitive deficits but for the past two weeks she has been intermittently disoriented to day, having visual hallucinations and experiencing periods of being “out of it,” during which she responds minimally to verbal stimuli. Which of the following diagnoses is lowest on your differential diagnoses for this clinical presentation?

- A. Alzheimer’s dementia
- B. Dementia with Lewy Bodies
- C. Subclinical seizures
- D. Delirium

The correct answer is A. The key feature of this history is that her symptoms are fluctuating. When you hear about acute onset of fluctuating mental status, we often think of delirium, but we must always rule out sub-clinical seizures and ask about other symptoms suggestive of Dementia with Lewy Bodies (i.e. acting out dreams, Parkinsonism). Although visual hallucinations and disorientation occur in Alzheimer’s dementia, onset is usually insidious and symptoms do not tend to fluctuate markedly.

BONUS MCQ (2025)

The 42-year-old daughter of a patient with vascular dementia asks you what she can do now to reduce her risk of developing vascular dementia herself. Based on the Lancet 2024 Commission on dementia, you recommend all of the following except:

- A. Sleep at least 7 hrs per night
- B. Quit smoking
- C. Exercise 150 minutes per week
- D. Treat depression

The correct answer is A. The 2024 Lancet Commission identifies 10 modifiable dementia risk factors in mid-life (hearing loss, high LDL-C, depression, TBI, physical inactivity, diabetes, smoking, HTN, obesity and EtOH >21 drinks/week). They did not think the weight of evidence was adequate regarding sleep deprivation. The 2020 Canadian guidelines DO recommend Tx of sleep apnea and avoidance of sleep deprivation to reduce dementia risk.

BONUS MCQ (2025)

79-year-old lady is referred to you for management of multimorbidity and polypharmacy. She has type 2 diabetes (A1c 7.5), hypertension, CABG 6 years ago, COPD on Trelegy and mild amnesic cognitive impairment. He takes metformin 500 mg PO BID, amlodipine 10 mg PO qd, chlorthalidone 25 mg PO qd, bisoprolol 2.5 mg PO qd, ASA 81 mg PO qd and risperidone 2 mg PO BID that was started four months ago during a hospitalization for urosepsis complicated by hyperactive delirium. Her delirium symptoms have resolved completely for the past three months, and she is managing well at home, independent in all IADLs. What do you suggest with respect to her risperidone?

- A. Stop risperidone without taper.
- B. Taper risperidone dose by 25% every day.
- C. Taper risperidone dose by 25% every two weeks.
- D. Cross-taper risperidone with quetiapine.

The correct answer is C. According to the 2018 Canadian "Deprescribing antipsychotics for behavioural and psychological symptoms of dementia and insomnia" guideline (Can Fam Phys 2018; Jan 64;17-21) antipsychotic dose should be tapered by 25-50% every 1-2 weeks to prevent withdrawal psychosis, anxiety, insomnia or agitation.

BONUS MCQ (2025)

An 82-year-old male with past medical history of hypertension, diabetes and benign prostatic obstruction is referred for assessment of memory problems. His daughter says that in the 2 years since her mother died, her father has stopped cooking for himself and has missed appointments and bill payments. She describes him as increasingly irritable and disinterested in spending time with his grandchildren. He continues to drive without issue and just passed a functional driving assessment his daughter insisted on. MoCA 25/30 (points lost on delayed recall but cueable, serial 7s). PHQ9 22/27. What is the most appropriate first treatment for his memory issues?

- A. Fluoxetine 10 mg PO daily
- B. Donepezil 5 mg PO daily
- C. Aripiprazole 2 mg PO daily
- D. Sertraline 25 mg PO daily

The correct answer is D. Patient has symptoms of depression, PHQ9 is highly suggestive of depression and he has deficits in attention and retrieval of words (vs. storage, which would be more suggestive of Alzheimer's dementia). Lack of driving impairment suggests that missed bills, appointments and decreased cooking may be related to apathy rather than cognitive impairment. Sertraline and duloxetine are first-line therapies in depression in older adults. Fluoxetine not recommended due to prolonged half-life, aripiprazole only indicated as adjunct if first line therapy fails. Donepezil would be reasonable if after effective treatment of mood symptoms, patient continues to have cognitive and functional impairment.

BONUS MCQ (2025)

84M brought to ED from home for assessment of confusion worse than baseline over past week. Past medical history significant for heart failure with preserved ejection fraction, stage III CKD, type 2 diabetes, Alzheimer's dementia and depression and chronic sciatic pain. His blister pack contains empagliflozin 10 mg PO daily, metformin 500 mg PO BID, rosuvastatin 5 mg PO daily, donepezil 5 mg PO daily, brexpiprazole 2 mg PO daily and duloxetine 60 mg PO daily. On exam GCS is 10, temperature 39.0 degrees, HR 120 bpm, increased rigidity in all extremities, slow biceps and patellar reflexes. All of the following are appropriate next steps, except for:

- A. Hold home brexpiprazole
- B. Administer cyproheptadine 12 mg PO
- C. Administer lorazepam 2 mg IV
- D. Hold home duloxetine

The correct answer is B. This patient has neuroleptic malignant syndrome, likely due to duloxetine inhibition of CYP2D6 resulting in supratherapeutic levels of brexpiprazole. Cyproheptadine is used to treat serotonin syndrome, which is less likely in this patient as they are hyporeflexive and rigid.

Bonus MCQ (2024)

82yo woman with moderate Alzheimer's dementia. MMSE 21/30.
What medication is indicated to slow rate of cognitive decline?

- A. Memantine
- B. Memantine + Donepezil
- C. Donepezil
- D. Atorvastatin

The correct answer is C. Memantine is indicated in moderate-severe dementia (MMSE 5-13). Donepezil is still the first line agent.

BONUS MCQ (2024)

82 man with Alzheimer's dementia unable to manage finances independently, driver's license suspended after at-fault accident, not able to prepare meals independently anymore. He has been taking rivastigmine for 6 months and there has been significant decline in appetite and weight loss 15lb, no improvement in cognition. Family wants to know what more can be done. What would be the next best step?

- A. Stop rivastigmine and start memantine
- B. Continue rivastigmine and add memantine
- C. Continue rivastigmine and start low dose ASA
- D. Stop rivastigmine and start atorvastatin

The correct answer is A. While ChEI are first line, pt should be monitored for side effects including weight loss and discontinued if they occur. Memantine is indicated in moderate-severe dementia and could be an option for this patient. Decrease in dose, alternative ChEI or trial of Rivastigmine patch could be considered depending on patient particularly if cognitive benefit.

Bonus MCQ 2026

An 86-year-old woman is assessed in clinic after her third fall in the past year. She reports one fall while gardening and two while walking indoors on level ground, she did not sustain any injuries. She lives in a house with her daughter.

Medical history: hypertension, type 2 diabetes, and osteoporosis.

Medications: ramipril, metformin, alendronate weekly, and a daily multivitamin containing 1,000 IU vitamin D and 1,000 mg calcium.

On exam: BP 132/70 mmHg, no orthostatic hypotension. Vision corrected with glasses. Timed Up and Go (TUG) test: 16 seconds. 25-OH vitamin D is 80 nmol/L.

Which of the following is the most evidence-based next step to reduce her future fall risk?

- A. Increase vitamin D supplementation to 2,000 IU daily
- B. Recommend hip protectors
- C. Discontinue alendronate and switch to denosumab
- D. Perform a medication review

The correct answer is D. The World Falls Guidelines strongly recommends that a medication review and appropriate deprescribing of Falls-Risk Increasing Drugs should be part of multidomain falls prevention interventions. Increasing vitamin D is not necessary if she is not deficient. Hip protectors and osteoporosis management are relevant for fracture prevention, with the former having evidence only in institutionalized settings and not in the community.

Bonus MCQ 2026

A 92-year-old woman is brought to the Emergency Department by her family for “strange behaviour” over the past 2 days. She is febrile and has a productive cough and chest x-ray findings suggestive of pneumonia. She forgets that her daughter visited her earlier that morning and tells you the year is 1994. She makes it from December to September after you prompt her to tell you the months of the year backwards. She says she sees her long-deceased husband in the room. Which symptom is the most helpful in distinguishing delirium from dementia?

- A. Forgets that her daughter visited her earlier that morning.
- B. Thinks that the year is 1994.
- C. Unable to say months of the year backwards.
- D. Visual hallucination of her deceased husband.

The correct answer is B. Inattention occurs in 98% of people with delirium and doesn't tend to occur until later phases of dementia. Short-term memory loss and disorientation occur in 91% and 78% of people with delirium but are also very common in Alzheimer's dementia. Hallucinations occur in 18% of people with delirium and 34-70% of people with Alzheimer's dementia.
1. Br J Psychiatry. 2014 Dec;205(6):478-85. 2. JAMA Neuro. 2005 Oct; 62(10): 1601-1608.

BONUS MCQ 2026

A 72-year-old man presents for evaluation of memory concerns. Over the past year, he has had increasing difficulty remembering recent conversations and misplacing objects, though he remains independent in activities of daily living. He manages his own finances and medications. Cognitive testing with MoCA reveals 25/30, with impaired delayed recall. Medical history includes hypertension and type 2 diabetes, both well controlled. He takes ramipril and metformin. Physical exam and basic labs are unremarkable.

Which of the following interventions has the best evidence for improving cognitive outcomes in this patient?

- A. Vitamin E supplementation
- B. Mind-body exercise program
- C. Intensive glucose control (HbA1c <6.0%)
- D. Donepezil
- E. ASA

The correct answer is B. This patient meets a diagnosis of mild cognitive impairment (MCI). All physical activity is recommended by the 5th ACCD/TD, in particular aerobic and resistance training, and promising evidence for mind-body exercise. Clin Psychol Rev. 2025 Aug;120:102604.